

SELF-DRIVING LAB OF PHOTOPOLYMER SYNTHESIS AND MECHANICAL TESTING FOR EXPEDITED MATERIALS DISCOVERY

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Marcus Rocco Fratarcangeli

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**SELF-DRIVING LAB OF PHOTOPOLYMER SYNTHESIS AND
MECHANICAL TESTING FOR EXPEDITED MATERIALS
DISCOVERY**

Approved by:

Dr. H. Jerry Qi, Advisor
School of Mechanical Engineering
Georgia Institute of Technology

Dr. Aaron Stebner
School of Mechanical Engineering
Georgia Institute of Technology

Dr. Rampi Ramprasad
School of Materials Science and Engineering
Georgia Institute of Technology

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LIST OF SYMBOLS AND ABBREVIATIONS

SDL	Self-driving Lab
AI	Artificial Intelligence
DLP	Digital Light Processing
MCS	Material Composition System
CTS	Curing Time/Kinetics System
ATT	Automated Tensile Testing System
CNN	Convolutional Neural Network
UTS	Ultimate Tensile Strength
PID	Proportional Integral Derivative
DoF	Degrees of Freedom
DoC	Degree of Cure/Conversion
T_g	Glass Transition Temperature

SUMMARY

Material discovery has gained significant research attention in recent years as it has the potential to revolutionize the development of new materials for targeted applications. However, material discovery typically relies on hundreds and thousands of experiments. Self-driving labs (SDLs) have been intensively researched in recent years as they can leverage autonomous high-throughput experimentation and characterization to enable efficient materials discovery. They can effectively replace a human researcher by working with machine learning algorithms as the “brain” and with robots for dexterous actions. In a common scenario, an SDL will create samples, test, and collect data. Then it will use an active learning algorithm to update the design for the next experiment. After many iterations, the design space will begin to fill and the algorithm will learn the relationship between factors and responses, enabling the discovery of an optimized material without human intervention. In an SDL, automated machines, systems, and sensors must be created to interface with the computer and robots to perform specific research tasks, such as material synthesis and sample testing/characterization. Although many SDLs have been developed in recent years, no SDL has been made for the discovery of 3D printable photopolymers with targeted mechanical properties.

This work has developed three important components of SDLs for synthesizing and testing photopolymers. The first component prepares photopolymer inks using a robotic arm, closed-loop liquid dispensing system for monomers/crosslinkers, and powder

dispenser for photoinitiators. The second component cures photopolymer inks with a custom rheological sensor to determine time, temperature, and speed of curing. The third and final component automatically performs tensile tests on tensile bars made from the previous photopolymer inks using a robotic arm, force sensor, and automated clamp for grasping the samples. Simultaneously, the tensile testing system utilizes cameras for anomaly detection with machine learning, and digital image correlation for detecting sample dimensions. There is also automated data processing and property extraction with an algorithm that calculates and stores material properties including modulus of elasticity, elongation, and tensile strength. The three components can precisely create mixed photopolymer inks with desired concentrations of monomers, estimate curing properties for determining printability, and reliably provide mechanical properties, respectively. This work paves the way for more efficient materials discovery for photopolymer-based 3D printing.

CHAPTER 1. INTRODUCTION

1.1 Material Discovery

Modern technologies demand more materials with diverse properties. The task of discovering new materials is a research goal that spans many fields and industries. The active learning method has shown promise in discovering new materials. However, active learning is heavily reliant on experimentation. For example, a previous work discovering photopolymers in a three-component design space created over 50 different combinations of monomers and tested their mechanical properties [1]. In total, over 700 manual mechanical tests were performed. If the component space were to be expanded by one monomer, the required mechanical tests would reach over 1000. While the data collected is helpful and can facilitate a model to learn the relationships between the components, the research budget is very high and may not be worth the cost. Creating a dataset large enough to gather any meaningful information about the properties would require an enormous upfront cost.

Therefore, different methods of accelerating the materials discovery task have been proposed [2-4]. For example, high throughput characterization methods have been shown to expedite the data collection aspect of materials discovery [5-8]. Hundreds of samples can be tested in hours, instead of days. In addition, methods to choose samples more efficiently to decrease the research budget show promise. Instead of relying on a researcher's intuition to choose the next sample, active learning can be employed. In some cases, a researcher's intuition will be sufficient. But as a design space expands and becomes

more complex, computerized algorithms start to shine. Combining high throughput processes and active learning can lead to a system for materials discovery that can effectively replace a human researcher after being implemented.

1.2 Self-Driving Labs

The invention of robots and artificial intelligence (AI) have enabled smarter and more efficient problem-solving over many industries and research areas. One of the major tasks affected by the combination of robotics and AI is materials discovery. Historically, materials were discovered with the trial-and-error method over months or years of time. While this method is still effective and has yielded many innovations, the advancement of intelligent materials discovery is beginning to demonstrate its effectiveness as a situational replacement.

Humans have two major functions that are difficult to replicate: dexterous action and decision-making. Dexterous action refers to tasks skillfully performed with the hands. For example, picking up a pipette and extracting liquid from a container. This task may seem trivial to an experienced researcher, but there is significant coordination, sensing, and communication required that would be very challenging to perform by a robot. In only picking up the pipette, eyesight is used to identify the pipette and coordinate a hand to grasp it with its fingers. A robot needed to perform this task requires an active vision system continuously feeding information to the software to direct the robotic hand. Decision-making is selecting an action while comparing it to other alternatives. If a researcher is deciding what his/her next experiment should be, he/she needs to consider all the

alternatives and the relative costs of those alternatives. In addition, previous experiments must be considered to predict what the resulting output from the new sample will be. For example, if a researcher makes one material with three different concentrations of three different components, and it yields property A, but property B is desired, the researcher now needs to revisit his/her material composition and decide how to make changes for the next iteration. The researcher can intuitively discover the relationships between each component and their respective concentrations, but it will take many iterations, which is neither cost nor time effective. This is the case where assistance from robots and AI would be most helpful.

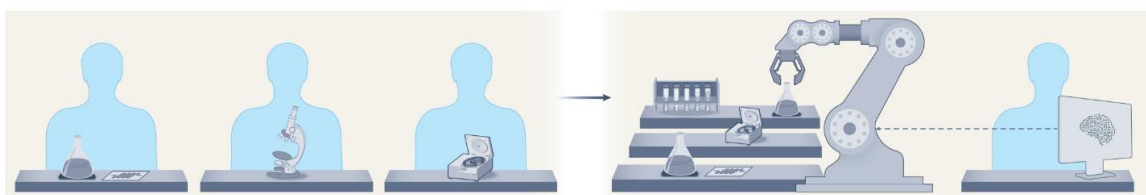


Figure 1: Depiction of self-driving lab theory [10]

The concept of a self-driving lab (SDL) is a combination of automated machines, robots, and decision-making algorithms that can perform the tasks of a researcher more efficiently to accelerate the accomplishment of research goals [9]. Figure 1 illustrates the definition of an SDL [10]. While they are not always called self-driving labs, the name encompasses all systems which have similar definitions. Further, SDLs can be broken down into categories based on their goals and autonomy, as well as hardware and software as shown in Figure 2 [9]. The lowest level of an SDL (Level 0), which may not be an SDL at all, is where the experiments and selections are all performed manually by the researcher. An intermediate level (Levels 1-4) would consist of some automation, perhaps a workflow or multiple experiments, and the experiment selection is being performed by a computer.

The highest level (Level 5) of an SDL is completely autonomous in its hardware and the search space, and experiment selection is generated by a computer. At Level 5, a user would be able to input a goal and let the system run until that goal is achieved, or materials need to be restocked. The SDL level typically depends on the research budget, goals, and utility of the system. It should be noted that not all lab or research tasks will benefit from the application of an SDL. On one end of the spectrum, research goals with very small design spaces or very inexpensive samples will likely not be worth the upfront cost of an SDL. The time and financial cost of creating an SDL may outweigh the utility, since that time can be spent experimenting. In addition, it is possible that some design spaces may be intuitive enough to not require complex software algorithms to find relationships. The hardware could benefit from automation to create samples very efficiently, but the relationships could be linear and easy to extract. On the other end of the spectrum, there

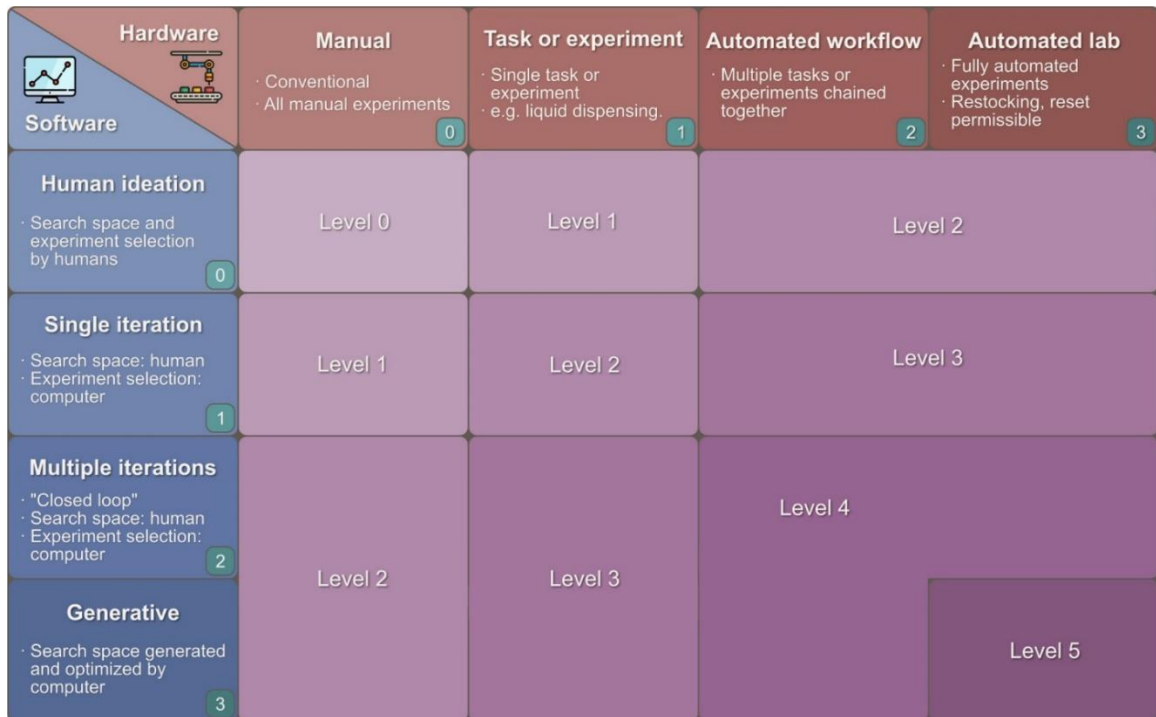


Figure 2: Various levels of an SDL [9]

could be some lab tasks or samples that are extremely expensive and the cost of automating those tasks is even more expensive, making it infeasible to implement. On top of that, the tasks could also be very low throughput, and even with automation, it would take years to collect enough data points to feed a complex algorithm. Everything between these two extremes is possible, and the acceleration of research goals is not always feasible.

The primary components of an SDL in terms of hardware and software are the processing, characterization, and decision-making systems. Using the levels of an SDL, the processing and characterization systems are typically on the hardware side, whereas the decision-making is the software. These systems will be discussed in more detail in the subsequent subsections.

1.2.1 Processing

To test a sample, a sample must be made. The first major component of SDLs is making, or processing. As most SDLs are created to replace a researcher performing physical tasks, there is usually a physical sample being made. The sample type is typically a liquid or solid, such as an inorganic powder, or monomers, as these are common lab

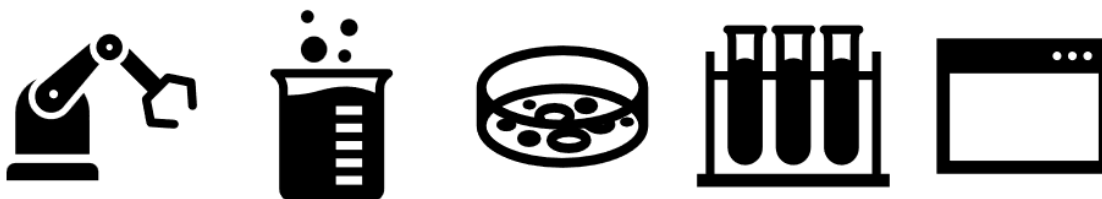


Figure 3. Common processing techniques, such as: robotic handling, curing, petri dish processing, chemical mixing, heating/cooling

materials and can be processed easily. The actual processing could consist of any number of procedures as shown in Figure 3; the primary example being mixing. However, once different components in a material are mixed, there could be further steps to transform that material into a usable format for characterization. For example, some materials are deposited onto glass slides, while others can be dispensed in a well-plate format. Processing is highly variable as different research goals have different requirements for their samples.

In any SDL, the processing system receives commands in some form from the decision-maker to create certain samples which it believes could further the research goal. In each iteration, the processing system will make the samples to use for characterization. It is critical that they are made properly and consistently. Inconsistencies in the process will lead to increased error and uncertainty with data collection, creating a less effective model and system overall. The nature of the material handling, like the powders or monomers, causes difficulties when automated as there is much variability between different powders or monomers and their behaviors. However, a researcher handling the raw materials manually still maintains a healthy margin of error in the sample preparation. Therefore, a comparison between the same task performed by the system and manually by a human is helpful to verify the accuracy and precision of the system.

1.2.2 Characterization

After the samples are successfully processed (or made), they must be characterized to collect the necessary data. Characterization methods can range drastically from rheological to spectroscopic and can be contact or non-contact, as shown in Figure 4 [11, 12]. Most existing characterization methods are non-contact based, such as imaging or spectroscopy. There are also simple contact methods, such as indentation, that do not require much sample manipulation. Many methods augment existing lab devices to enable their automation, while others create entirely new devices for their specific purposes. An example of characterization for photopolymers is uniaxial tensile testing. A single tensile test can yield many mechanical properties of the sample to help characterize and compare to other samples. A uniaxial testing machine set up for compression testing does not need much adjustment. The sample can be placed autonomously onto the test platform, the test runs, data is collected, the platform is cleaned, and the process repeats [13]. This is an ideal case. A more complex case would be the same uniaxial testing machine, but for tensile testing. Here, the sample needs to be secured properly in the two grips of the machine without human intervention, which is a much less trivial task. An autonomous tensile testing system has not been reported in literature and is one of the primary contributions of this work. Tensile tests are sometimes desirable because some mechanical properties, such



Figure 4. Common characterization techniques, such as: imaging/spectroscopy, microscopic imaging, thermal testing, weighing, and measuring

as tensile strength and maximum tensile strain, can only be measured by tensile tests.

Once the samples have been processed and characterized, there is raw data corresponding to each sample that will be analyzed by the decision-making algorithm. Before that step, the data needs to be preprocessed autonomously. Commonly overlooked, autonomous data processing can be a challenging task as it requires algorithms that are robust enough to handle any cases and deal with errors while reliably preparing the data correctly. An example is the tensile testing data analysis. A tensile system typically outputs force and position data, from which stress and strain can be calculated with the addition of the sample cross-sectional area. In addition, to find a specific property like the modulus of elasticity, which can vary drastically between samples, the analysis must be able to consistently identify the correct modulus. If it cannot, the sample will become worthless unless a researcher manually adjusts the analysis to include that sample. In a worst-case scenario, the property value is found incorrectly and fed to the decision-making algorithm, which then makes poorer decisions as it is learning from incorrect data.

1.2.3 Decision-Making

The decision-making process is perhaps the most critical in an SDL and its importance increases proportionally to the size of the design space and complexity of the research goals. Larger design spaces will create difficulty for humans to make effective decisions about the data, and statistical methods may even be too simple to capture some relationships. Therefore, a computer-based decision-maker becomes necessary. Typically, this takes the form of an active learning algorithm, which is a type of machine learning

algorithm that updates its prediction as new data becomes available. There are two primary types of active learning methods: each time new data is acquired, 1) it is added to the existing dataset and the model retrains on the entire dataset; or 2) the model is incrementally updated with new data, which is based on Bayesian approaches [14, 15]. The loops for each type are shown in Figure 5.

Both methods can be seen in research with success, and they are situation dependent. For example, if each iteration yields only one sample and the research goal is a certain mechanical property, then a Bayesian approach would be wise as each iteration the design will change based on the input from the one sample, eventually leading to the goal. Whereas for an iteration that yields result from many different samples (such as 10s or 100s) and the research goal is discovering new material, the first approach would be best as it can expand its dataset and find new relationships from all the existing data.

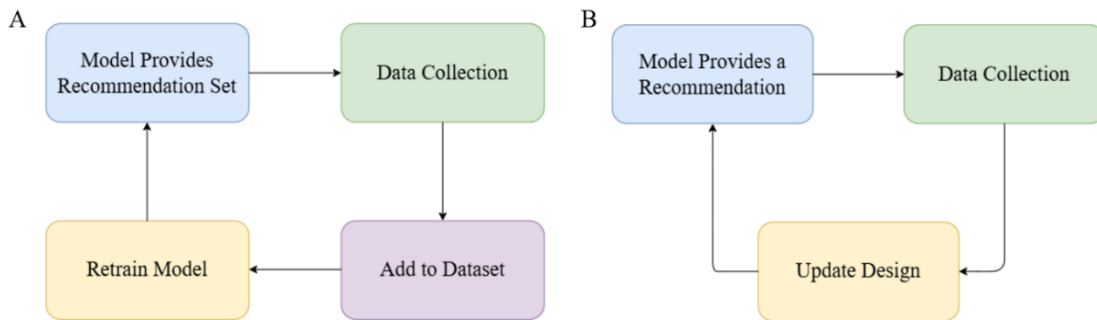


Figure 5: Two main types of active learning. A) batch update and B) incremental update

Decision-makers also need a starting point. In most cases, this involves design space search algorithms, like Latin Hypercube [16], that will sample the space evenly and create an initial dataset. From here, the model can begin to gather some initial relationships

and make decisions with relatively high uncertainty. As the SDL runs, more data is acquired, and the model can start making more precise predictions.

1.3 Motivation and Objective

The areas of study in this thesis are the development of an SDL for photopolymers and the application of the lab to discover new photopolymers with active learning. Photopolymers are widely used in industry for coating, dental, and 3D printing applications. For practical applications, a photopolymer consists of multiple monomers to enable tuning of the properties, as demonstrated by Yue et. al, with an example shown in Figure 6 [17]. SDLs have enabled the discovery of new materials with complex properties in large design spaces, and the objective is to leverage this technology for photopolymers [18-20]. Specifically, photopolymers are combinations of monomers and/or crosslinkers used for digital light processing (DLP) 3D printing. There are an unlimited number of combinations with drastically different properties. Manually processing and testing photopolymer samples is time consuming, and the design space is too large to manually

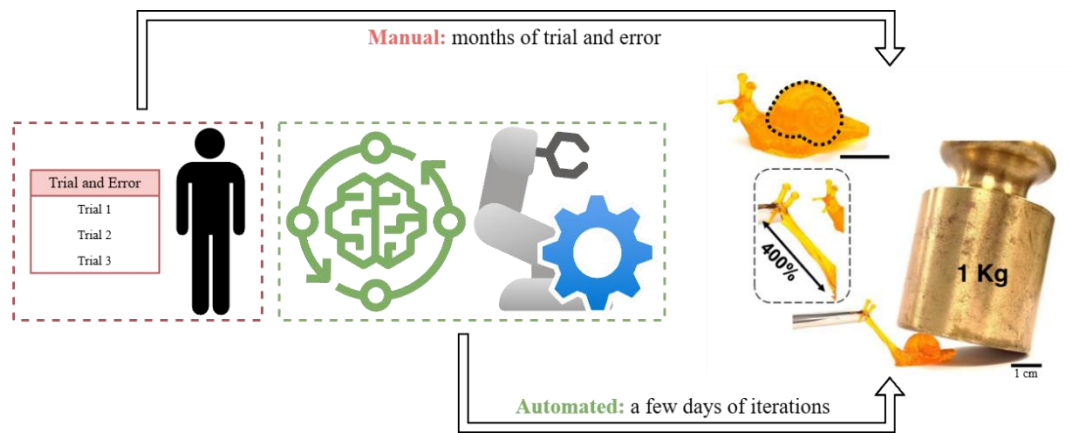


Figure 6: The discovery of a unique photopolymer can be accelerated with an SDL

understand the relationships between many different components. In addition, the cost of photopolymer samples is relatively low, the throughput is high enough to facilitate efficient data collection, and the time saved with an SDL is very high. Therefore, this problem is an ideal application for an SDL.

1.4 Outline

The thesis is divided into three chapters. Chapter I introduces SDLs and their components, along with common practices and current methods. Problems with making and testing photopolymers are proposed, and the need for an SDL is examined.

Chapter II discusses the physical system. Each component of the SDL will be reviewed and its importance to the overall system explained. The integration of all the components to form the overall system is also discussed.

Chapter III exemplifies the pairing of physical SDL with an active learning decision-maker to discover new photopolymers through a specific use-case.

Chapter IV summarizes the work from this thesis and provides areas for future work.

CHAPTER 2. SELF-DRIVING LAB SYSTEM DESIGN

The design, development, and verification of the physical components of the SDL are discussed in this chapter. The SDL consists of a material processing system, a curing time system, and a uniaxial tensile testing system, as shown in Figure 7. These components are automated to perform their tasks with minimal, if any, human intervention. The time to create, process, and test samples is greatly reduced through this hardware and enables the researchers to perform other tasks while the systems run.

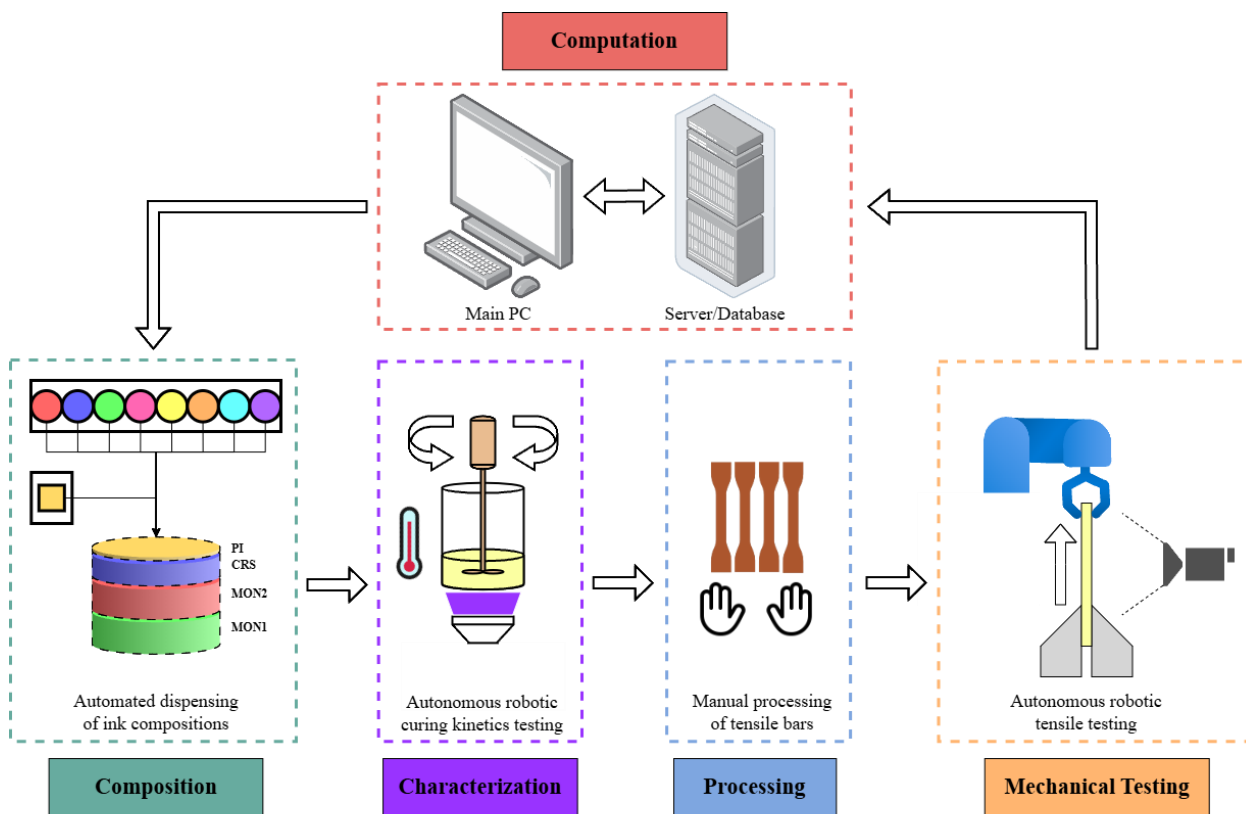


Figure 7: SDL schematic showing the main systems

2.1 Layout and Design

The entire SDL is laid out on a custom-made modular table (Vention, Montreal, CA, USA) that allows for hardware mounting anywhere with extruded aluminum bars and t-slots. Figure 8 shows a depiction of the SDL table and where each system is located, Figure 9 is an actual image of the entire SDL in the laboratory, and Figure 10 is the top view, showing the real arrangement of different components. The two robots are mounted at either end of the table and face the center, providing them with the most reach and largest work area possible. The right robot (Robot 1) manages the material composition system and is surrounded by other components: the container dispenser, the cork dispenser, the liquid dispensing system, the powder dispensing system, and the magnetic mixer. The left

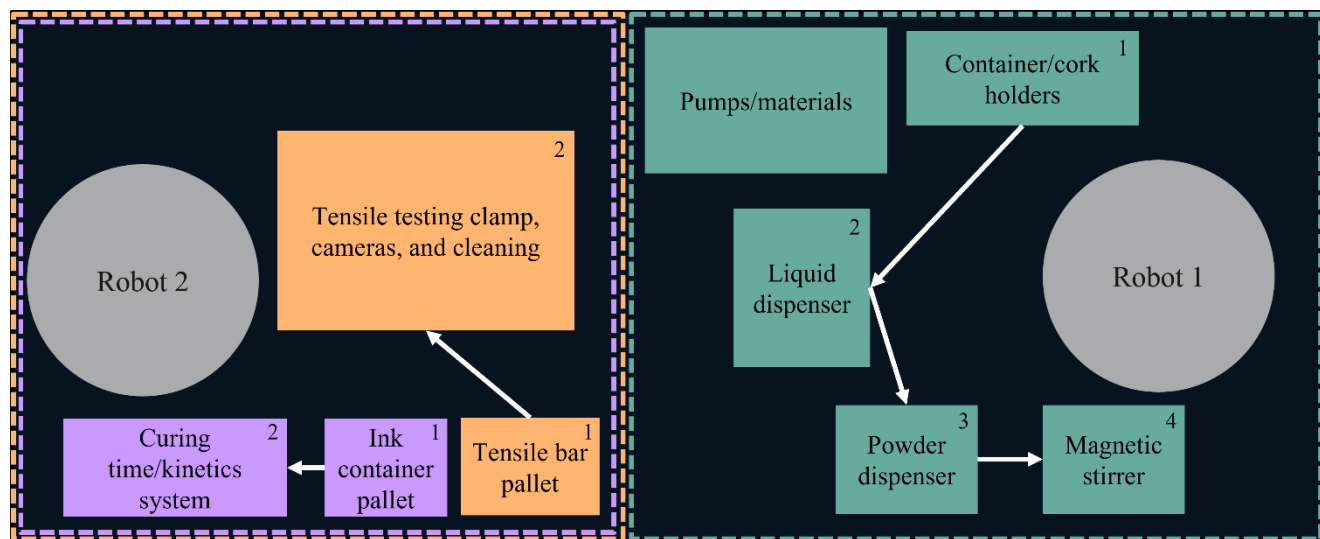


Figure 8: Top-down schematic of photopolymer SDL mounted to table



Figure 9: Real image of the table in the lab space

robot (Robot 2) is responsible for the automated characterization systems. Directly in front of it is the automated clamp and cameras, then the cleaning stations for the grippers, and to the right is the curing time testing system. Having the systems mounted to the same table allows them to potentially work together in future work. Additions can easily be made to the table to support an additional robot or subsystem to facilitate a completely autonomous SDL.

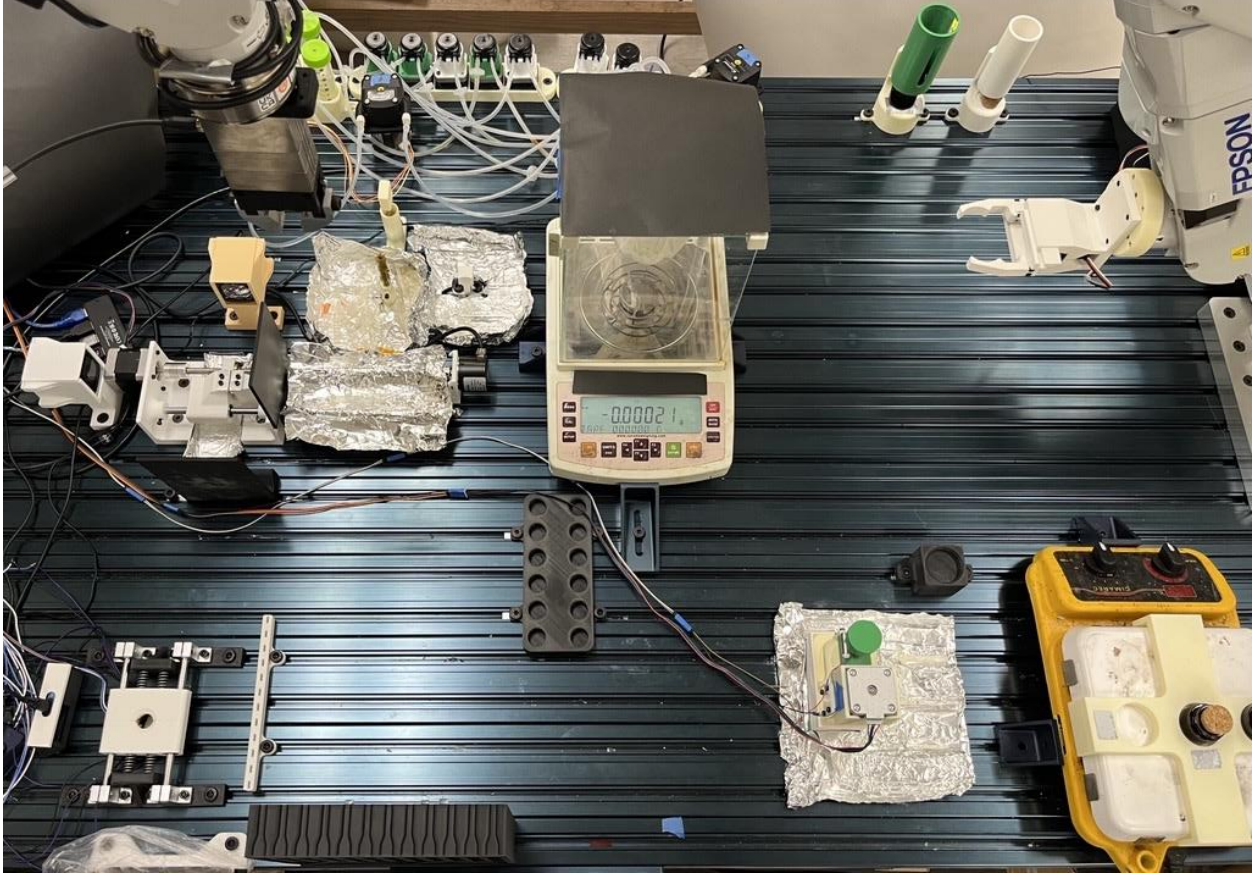


Figure 10: Real image of SDL mounted to table

2.2 Robotics

The replacement for dexterous action in the photopolymer SDL is Epson C8L robots (Epson America, Los Alamitos, CA, USA). They are programmed with the proprietary SPEL+ language. Each robot has a controller that is capable of handling I/O inputs and outputs, which is the method used for automation here. The original purpose of the robots is industrial automation, so they are lacking in ease of programming and integration, whereas cobots would be more effective. However, the strength and precision of the industrial robots is useful for the characterization techniques. The robots are programmed through the Epson Robot Controller software and the programs are uploaded to the robot controllers.

2.3 Software and Firmware

The SDL software is organized in a GitHub repository where the active learning model, dataset, and system control files are all present. More about the integration with active learning is discussed in the next chapter. The system control is composed of a set of Python programs which dictate all actions performed by any of the sub-systems. The pipeline starts with the Python program which is connected serially to several Arduino boards. When the robot must perform a task, the Python code sends a serial phrase to the corresponding Arduino. The Arduino is in a constant state of checking for signals, and when received, turns on a relay which sends an I/O signal to the robot controller. When in the active “run” state, the robots wait for I/O signals and perform the corresponding tasks when received. Once the robot task is complete, the robot outputs an I/O signal to the Arduino, which is forwarded onto the Python code. The feedback loop is necessary to ensure the status of each system is tracked as many systems must work simultaneously and in certain orders. During the robot task, the Python code is in a waiting state for the signal from the robot before it proceeds to the next step in the code.

The three components, material composition system (MCS), curing time system (CTS), and automated tensile testing (ATT), have their own respective software and firmware, but are all managed by the driver in the repository. A single iteration would work as follows: the driver is run, and the active learning model chooses compositions of monomers and crosslinkers to send to the material composition system. The MCS is then commanded to begin, and the inks are mixed and prepared. The driver then pauses and waits for a user input before proceeding to characterization. Once the inks are mixed, a

researcher must manually prepare the vials and the tensile bars for both the CTS and ATT systems. Referring to the levels of an SDL in Chapter I, this SDL is a level 4 system. The breakdown in the completely autonomous loop is the manual processing of vials and tensile bars for the characterization systems. Once these are prepared, the CTS will commence and test all the inks. The driver then waits for another user input before proceeding to the ATT. This is because the two systems use the same robot and cannot be run simultaneously. Also, the tensile bar samples must be placed in a pallet before being tested. Then the ATT proceeds and all the samples are mechanically tested. The data from the CTS and the ATT are stored in CSV files and merged before being sent to the server.

2.4 Material Composition System (MCS)

The photopolymer processing system contains two main components: the liquid dispenser and the powder dispenser. In addition, there are stations, or pallets, that hold new containers with magnetic stirrer bars, caps for these containers, and a magnetic mixer for the finished inks.

2.4.1 Photopolymer Processing

The liquid dispenser is the primary component of the photopolymer processing subsystem. The raw materials in this case are liquid monomers and therefore can be dispensed to make the composition for testing. The raw monomers are held in vials which have tubes connected to peristaltic pumps. The pumps are stepper motors with bespoke

peristaltic mechanisms attached. Stepper motors are used instead of traditional DC motors as their speed can be controlled and adjusted easily. A proportional integral derivative (PID) control algorithm is used to command the pumps to ensure accuracy and precision of the dispensed liquid. There is an Arduino Mega (Arduino, Italy) microcontroller that receives commands from the Python code on the PC to control the pumps. The speed of the pumps (steps/sec) is actively adjusted as the dispensed mass nears the goal.

The empty containers are placed on a scale with the pumps exit directly above. The mass is measured continuously as the liquid is being dispensed. Figure 11 schematically shows the closed loop control circuit for the liquid dispensing. In most cases the weight for a composition is 8g, but since most compositions contain three or more components, any combination to reach 8g total can be used. Therefore, to verify the precision and accuracy of the system, inks of various amounts were dispensed and their actual mass compared to

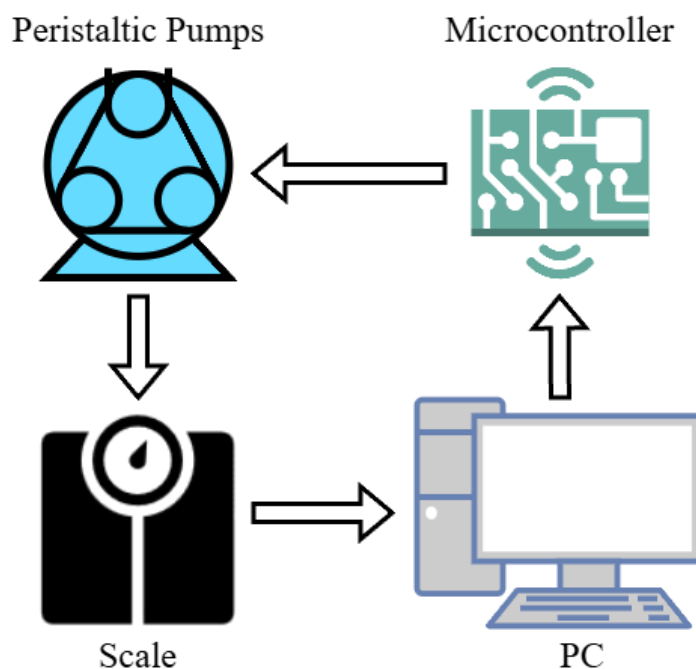


Figure 11. Closed loop liquid dispensing circuit

the expected mass Figure 12 shows the residual plot depicting the dispensing accuracy and precision for various amounts.

As shown in Figure 12, the percentage error is very low for the dispenser because of the PID control algorithm. When the dispensed mass nears the target, the speed slows down very significantly. This is done for two reasons: it allows the scale time to send the mass as there is delay in the signal and it ensures that there is no overshoot because of the delay. There is a tradeoff with the time it takes to dispense and the accuracy, but it is insignificant given that the system runs without human intervention. The resolution of the dispensing system is very low because of the nozzles used for dispensing. The silicon tubing to transfer the monomers through the pumps has an internal diameter of 3 mm. At

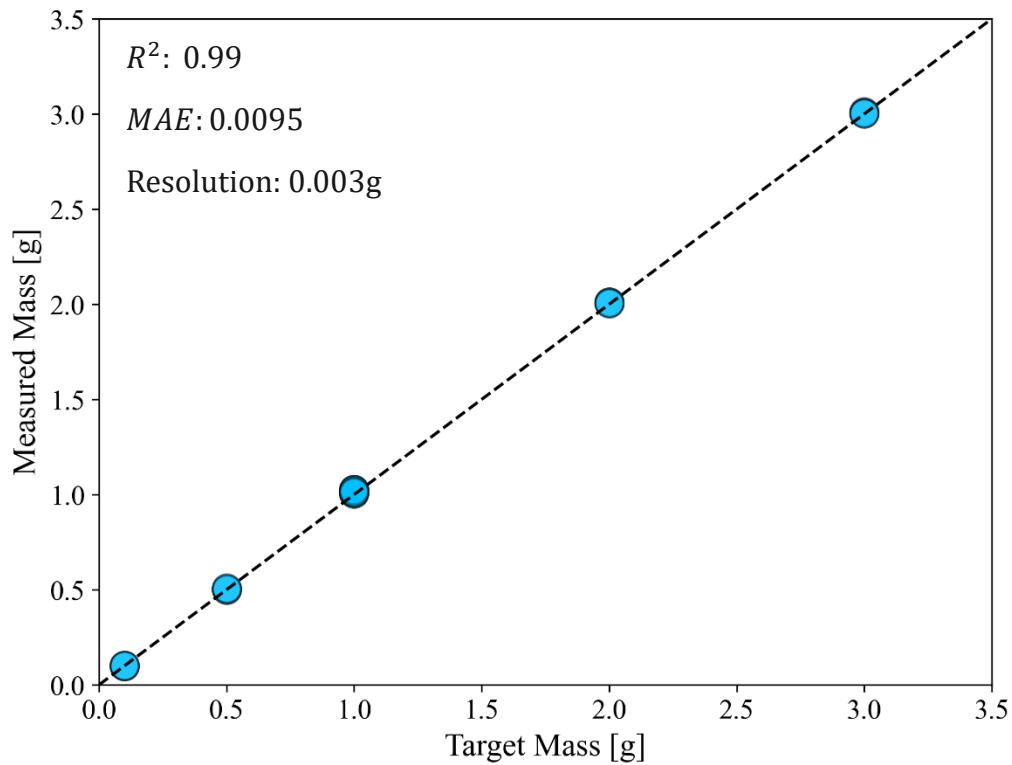


Figure 12: Residual plot for liquid dispensing system

the end of each tube, there is an 18-gauge (0.838 mm) nozzle that decreases the diameter significantly. Each droplet of liquid shrinks to around 0.003 grams because of the nozzle. It is important to note that the viscosity of liquids being dispensed here is relatively low. Their viscosities are near water, or slightly above, and are easy to pump and dispense. If the system were to be expanded to encompass more viscous components, the PID algorithm must be adjusted to account for the change. Also, a heating system might be added to decrease the viscosity of these monomers.

2.4.2 Powder Dispenser

The powder dispenser is used to dispense photoinitiator into the inks after the monomers and crosslinkers are dispensed. The device consists of a stepper motor, a hopper, and flat sprockets, and is controlled with an Arduino Mega (Arduino, Italy). The physical

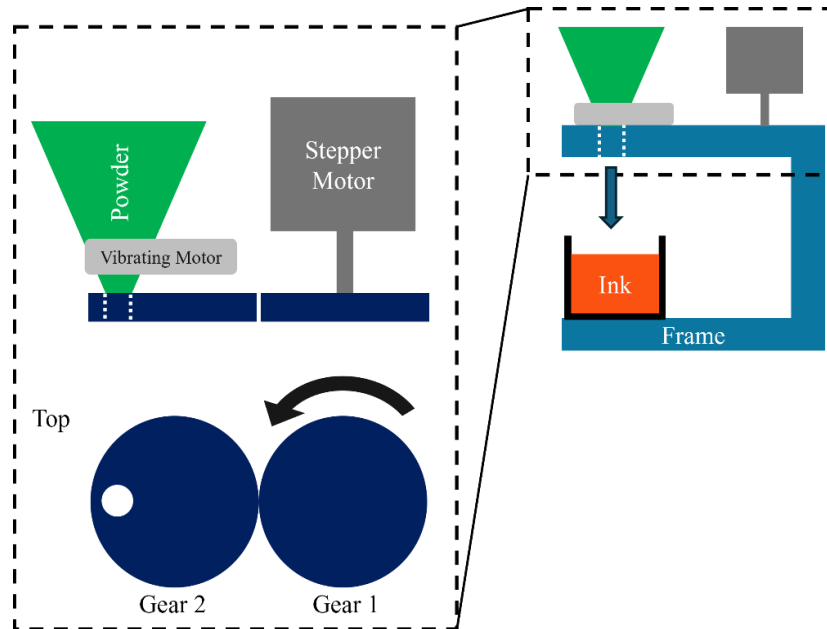


Figure 13: Depiction of powder dispenser hardware

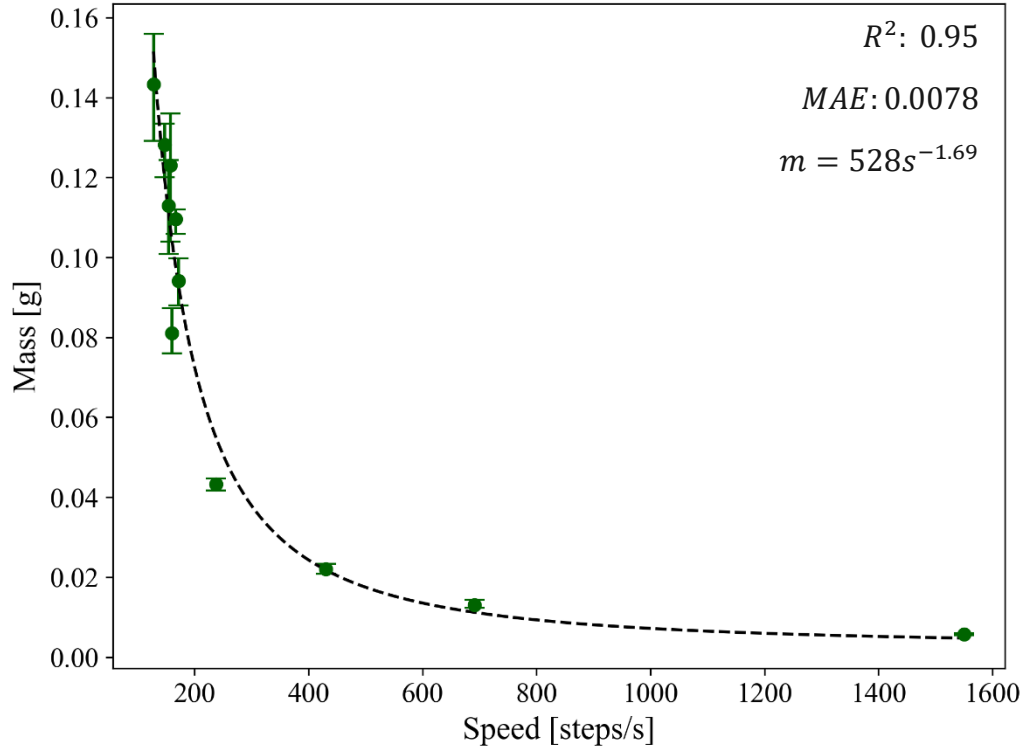


Figure 14: Dispensed mass curve fit

depiction of the hardware and operating mechanism is shown in Figure 13. The hopper contains the photoinitiator and sits above the second sprocket, which has a through hole on its face. When the stepper motor turns the first sprocket, the hole revolves and passes underneath the hopper, which allows powder to fall through. The amount of powder dispensed is determined by the stepper motor speed. In addition, a small vibrating motor is in direct contact with the hopper and vibrates when the device runs to allow the powder to fall cleanly and consistently, not sticking or clumping.

Commands are accepted through the Arduino as mass values. Through experimentation, a fit line was developed to model the relationship between the speed at which the gears turn, and the amount of powder dispensed, which is illustrated in Figure 14. The typical amount of photoinitiator used in any ink composition for DLP is 1%.

Therefore, for a normal ink mass of 8g, 0.08g of photoinitiator is desirable. The error in this mass is approximately $\pm 5\%$. However, as shown in Figure 14, the error decreases significantly as the speed increases. Therefore, the system sprockets can be spun multiple revolutions to allow for more accurate dispensing. For example, at approximately 400 steps/s, 0.02g can be deposited with less error (3.8%), and 4 revolutions can fulfill the task.

2.4.3 *Other Hardware*

The three remaining pieces of hardware in the processing system are the container dispenser, cork dispenser, and magnetic mixer. The container dispenser is the first stop during an iteration of processing. The robot will grab a clean container and bring it to the liquid dispenser to receive the composition. The container also contains a magnetic stirrer bar for when the mixture reaches the magnetic mixer. After the container has collected both monomers/crosslinkers and photoinitiator, the cork is placed on top to ensure that no light penetrates the ink and that no ink spills during mixing. Once the cork is secured, the container is placed onto the magnetic mixer where it will stir for 15 minutes. Since the inks are not viscous, they mix very quickly and do not require a long mixing process. Once the inks are completely mixed, they are removed from the magnetic mixer and placed on the table where they can be picked up by a user to prepare them for characterization.

2.5 Curing Time System (CTS)

The first system for the characterization of photopolymers is for curing kinetics. Before physical samples can be made, there must be a method to verify the crosslinking ability of the inks to form cured solids. Further, there are several important factors within curing kinetics, such as curing time, curing temperature, and the realizability. These parameters must fall within certain thresholds for the ink to be considered usable for DLP printing. Therefore, the curing kinetics system is used to gather this data for each ink and determine its viability. The hardware consists of a robotic arm with a force (torque) sensor and gripper which places a probe into the photopolymer ink. The data collection process begins with the robot wrist joint rotating slowly and turning the probe. At the same time, the thermal camera turns on and begins collecting data. The UV LED lights then turn on to begin the reaction. Based on free-radical polymerization kinetics, it is assumed that the polymerization reaction is exothermic. Therefore, as the resin is cured, it is expected that the temperature will rise. Also, when the resin becomes solid, it is expected for the torque required to turn the probe to increase significantly.

Based on conservation of internal energy the temperature change can be related to the change in monomer concentration [21].

$$\rho C_p \frac{\partial T}{\partial t} = \kappa \nabla^2 T + \frac{d[M]}{dt} \Delta H$$

where the rate of monomer consumption can be shown as

$$\frac{d[M]}{dt} = -k_p [R][M]$$

where ρ is the density, C_p is the thermal capacity, κ is the thermal conductance, and ΔH is the total enthalpy released. As bonds are formed during the polymerization, the temperature increases as heat is released. The temperature increase proves that a reaction is occurring, but it does not ensure there is a crosslinked network being formed. In some cases, for example, a monomer alone in an ink, may react and form bonds, but the ink itself remains a liquid because a solid network is not formed. Therefore, the hardness (or stiffness) of the ink can be measured with a probe during the reaction to guarantee a solid is formed. A combination of both the temperature and hardness of the ink during the reaction can help discover basic curing kinetics.

2.5.1 Curing Station

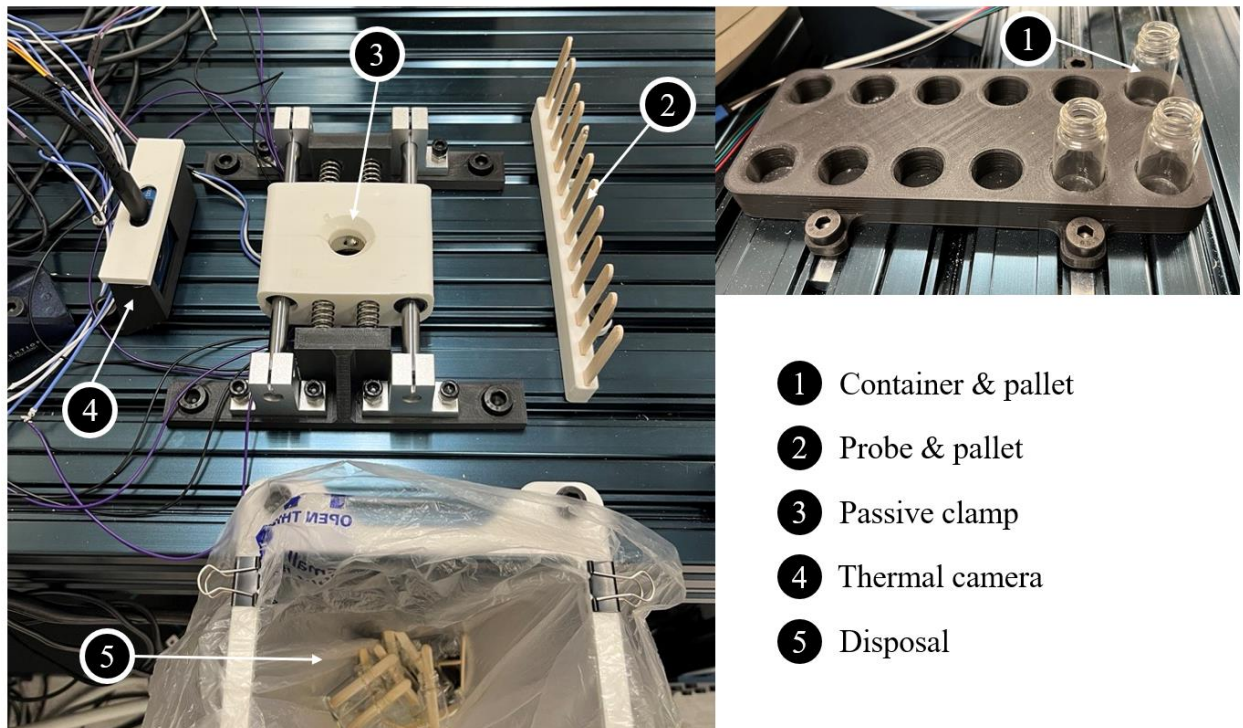


Figure 15: Curing time system hardware

The curing station hardware consists of the pallet which holds the inks, the pallet holding the probes, the passive clamp where the test occurs, the wireless thermal camera, and the disposal bin for completed test inks. Figure 15 shows the hardware systems and their orientation on the table. Each ink is dispensed into a clear glass container, for which 0.5g of ink is contained. The probes are small wooden sticks. A set of 12 tests can be performed in series without human intervention given the capacity limit of the pallets.

The robot being used for curing kinetics testing is the same robot for the tensile testing system. The details of the robot and end effector are discussed in more detail in the

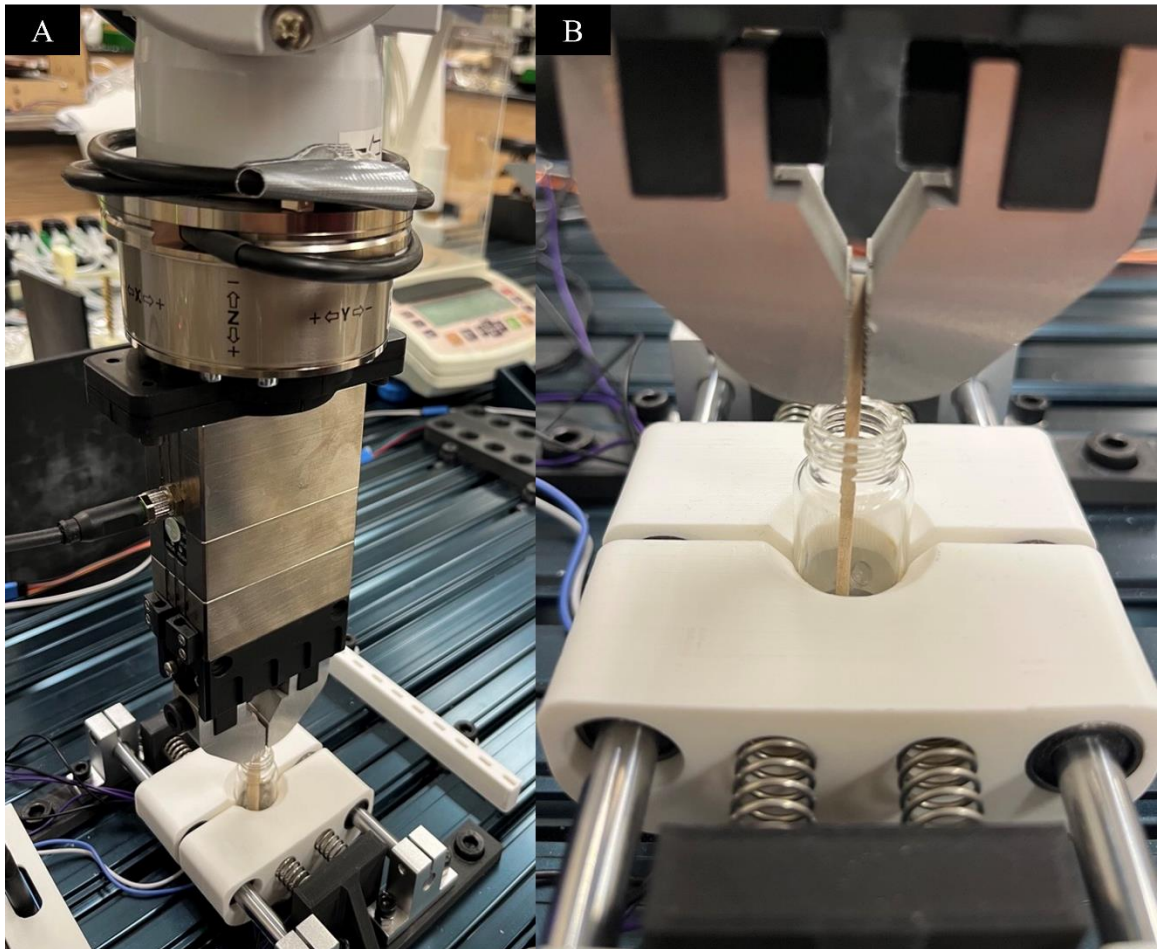


Figure 16: Robot with mounted force sensor and gripper in the curing time system. A) the robot wrist, force sensor, and gripper and B) the gripper holding a probe in the container for a test

following sections. Briefly, the robot has a force sensor with torque sensing capabilities as well as a servo gripper mounted on the end, shown in Figure 16.

A test begins with the robot picking a glass container from the pallet and bringing it to the passive clamp. The clamp is not powered, and the compression force comes from springs. There are two hands that are pushed together by the springs, and each has a half-circle cutout where the container fits into. There is a chamfer at the top of each half-circle which allows the robot to press the container into the slot, pushing the hands to either side. Also, the hands are fabricated from ABS plastic, ensuring a resistance to the temperature output of each ink. Below the container and circling the bottom, there are UV LEDs, with the strongest and most significant LED array being on the bottom directly underneath the container. Once the container is in place, the robot will grab a probe from the pallet and place it inside the container, submerging into a small layer of ink. The probe does not touch the bottom of the container but is as close as possible considering the limitations of the robot. There is an estimated 0.1mm distance between the container surface and probe.

The thermal camera is connected to a Raspberry Pi which controls the data gathering and conversion from pixel values to actual temperature data. Further, the Raspberry Pi and PC are connected wirelessly in a server/client relationship. The Pi acts as the server and accepts commands from the PC to start and stop collecting data. Once the temperature data collection is complete, the Pi sends the data back to the PC for analysis and waits to accept another connection or command. In addition, the robot force sensor is collecting torque data throughout the test. The first 30 seconds of the test are without any UV light to establish a baseline for the viscosity of the ink. After the light turns on, the ink

is expected to be reacting and hardening, and the temperature and torque will increase over time. The test ends either by the torque limit being reached on the robot, or the timer of three minutes running out. The torque limit is there as a safeguard but has never been reached by an ink in preliminary tests. The torques being measured are in the N-mm range and are insignificant when considering the force sensor is tested and characterized with torques in the N-m range by the manufacturer. The timer is present because the robot data collection system is limited to three minutes of continuous writing to a file. Either way, when the test ends, the robot drops the probe, grabs the container, extracts it from the clamp, and places it into the disposal bin. This process repeats for subsequent samples.

2.5.2 Data analysis

The output data from the physical test are the torque and temperature, both dependent on time. Initially, before any analysis is performed, there is compensation that must be made to ensure accurate temperature data. There is some inherent heat generation from the UV LEDs shining onto the container without ink. A series of dry tests without ink were performed to gather a baseline curve for the temperature due to the LEDs. The arithmetic mean was calculated over these tests and subtracted out from the measured temperature value during a real test. For the torque, the data is collected as the robot turns the wrist joint and therefore turns the gripper and probe. Attached to the side of the gripper, there is a wire that provides communication and power. While turning, the gripper wire causes a small torque on the gripper. However, the absolute torque values are not necessary and this can be ignored.

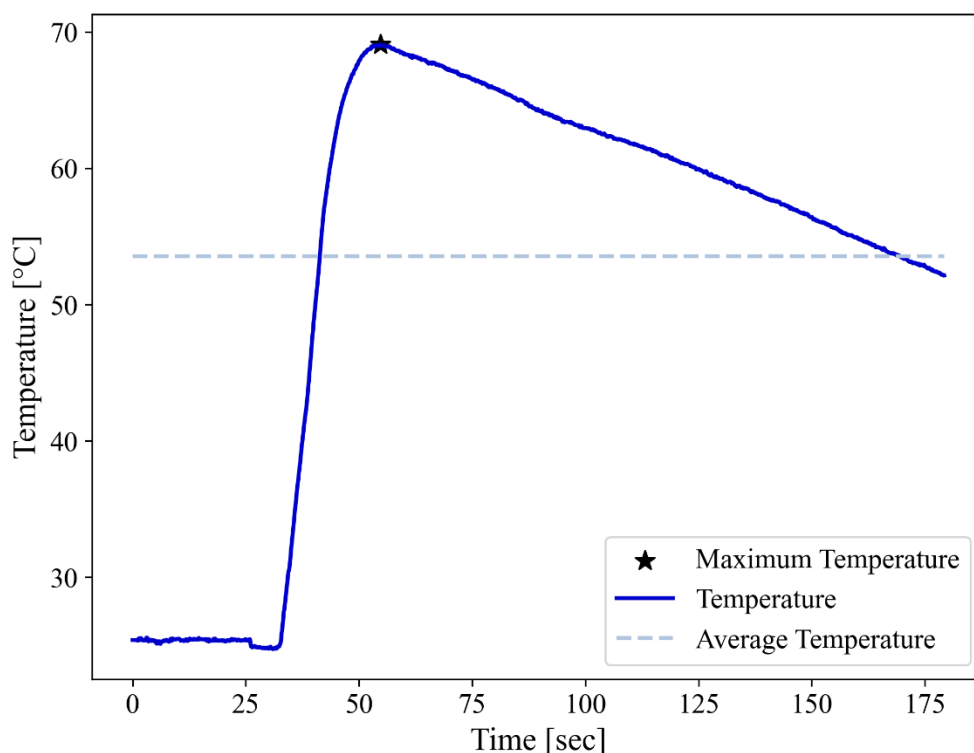


Figure 17: Temperature data measured over the duration of a cure test

Once the baseline curve for temperature was established, the following data analysis can proceed. The temperature data is analyzed first as it can assist in determining whether the ink cured or not before finding any additional properties. During the free-radical polymerization reaction, the energy output from the UV LEDs causes the photoinitiators in the ink to decompose, creating free radicals and allowing them to attack the bonds of the ink components, such as monomers or crosslinkers. The reaction is net exothermic because while some bonds are being broken, more bonds are being created from the formation of the polymer network. Therefore, the output energy can be measured in terms of temperature and show the progress of the overall reaction. As the reaction begins, the temperature typically increases sharply and steadily until the slope starts to decrease, eventually peaks, and slowly decreases as the reaction slows and stops, and the ink cools. Figure 17 shows an example of a temperature curve from a curing time test. The

first step in the analysis is to extract the maximum temperature and the time at the maximum temperature. The time can be used to discover whether the ink was still cured at the end of the test, which can help determine the realizability. In addition, the average temperature across the entire test is calculated. Occasionally, this can provide helpful information about the reaction. For example, if the maximum temperature is high but the average temperature is abnormally low, the reaction is likely to take a long time to start or progress very slowly. While this information is not one of the data outputs for active learning, it can still be helpful for understanding the inks.

Torque data is the primary source of data and is used to find the cure time and determine the true realizability. A key difference between polymerization reactions is the presence of crosslinking to form polymer networks. Crosslinking is critical to forming a solid structure because it limits the movement of the chains significantly. For example, an ink without crosslinkers would simply form long chains without forming a network and

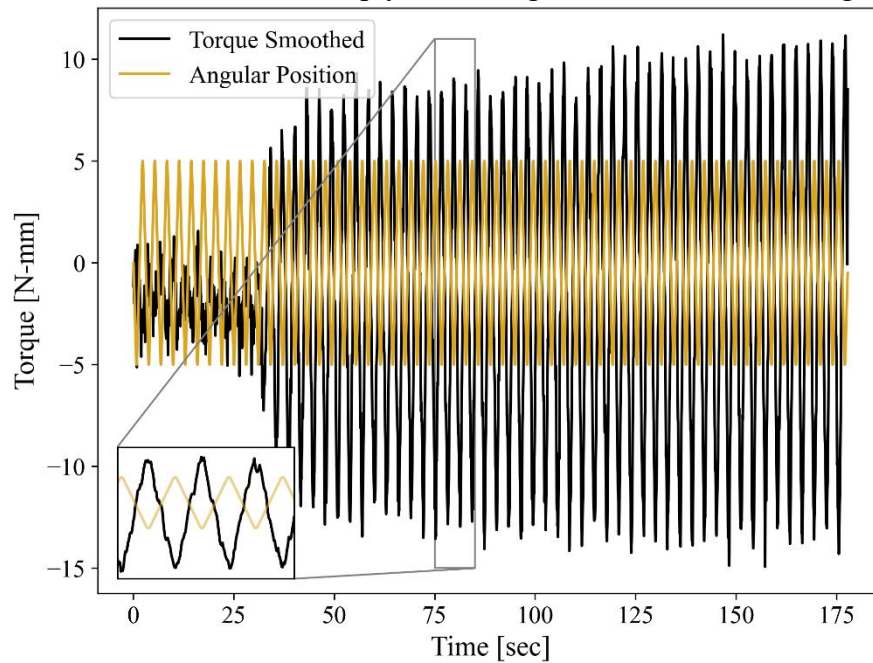


Figure 18: Raw torque and angular position over the duration of a curing test

would thus end up a viscous liquid instead of a solid. The polymerization reaction is still occurring, and the temperature would increase as bonds are still forming, despite no crosslinking. From the temperature data alone, it would appear as if this ink with only monomers has successfully cured, but it is still liquid. Therefore, the torque of the probe turning against the ink is used to verify whether the ink cured or not and find the primary properties. Figure 18 gives an example of raw torque data from a curing time test. The same sample is used as an example for all the data, including the temperature. The data output from the sensor is oscillatory and centered around zero. Any deviations causing the raw torque data to not look centered is because a small filtering algorithm is applied. Here, the Savitzky-Golay filter is used to smooth the curve. The angular rotational position is tracked and the torque at each peak is extracted. The probe turns ± 5 degrees in either direction and the peaks and valleys of the raw torque curve correspond to the extrema of the angular position. The torque values at peaks and valleys are then paired and the

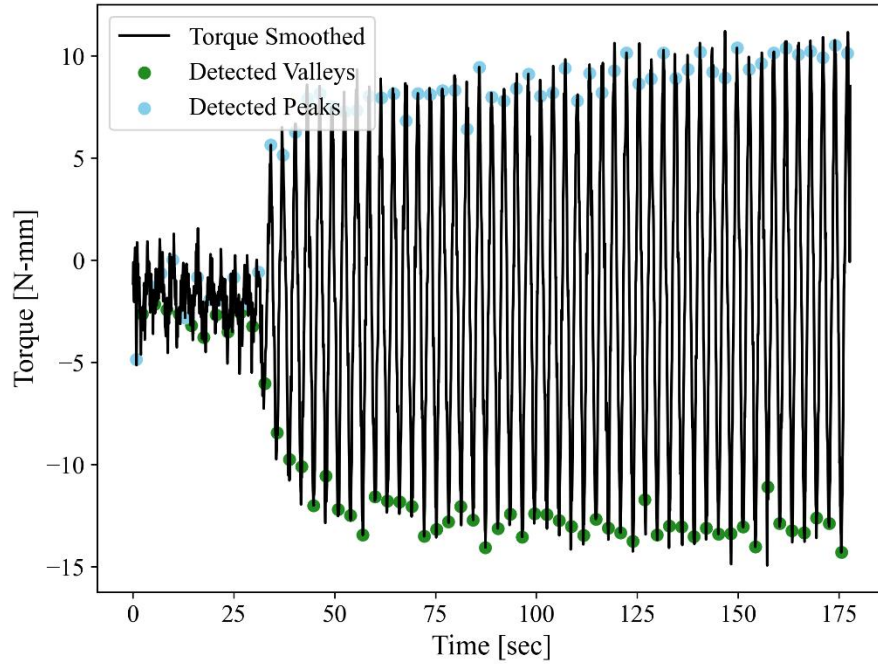


Figure 19: Raw torque data with valleys and peaks detected

difference is found to discover the amplitude of the torque. The detected peaks and valleys of the raw torque curve are labeled in Figure 19.

The amplitude values are normalized, and a sigmoidal curve is fit to them, as depicted in Figure 20. Ideally, during the time when the light is off, the amplitudes should be near zero, and when the light turns on, the ink will harden, and the amplitudes will increase gradually as the ink cures. Once the ink is fully cured, the torque amplitudes should remain constant at that value, providing a clear sigmoid curve. The curve fit is determined by the equation below.

$$f = \frac{A}{1 + \exp\left(\frac{x-h}{s}\right)} + c$$

where A is the parameter for the height of the top line relative to the bottom line, c is for the height of the bottom line, h represents the inflection point, and s is the distance between the top and bottom lines along the abscissa. It should be noted that the parameters can differ and still output the same curve shape. For example, the sign on A and s can be reversed, and c adjusted according to the change, and the curve will remain the same despite having technically different parameters.

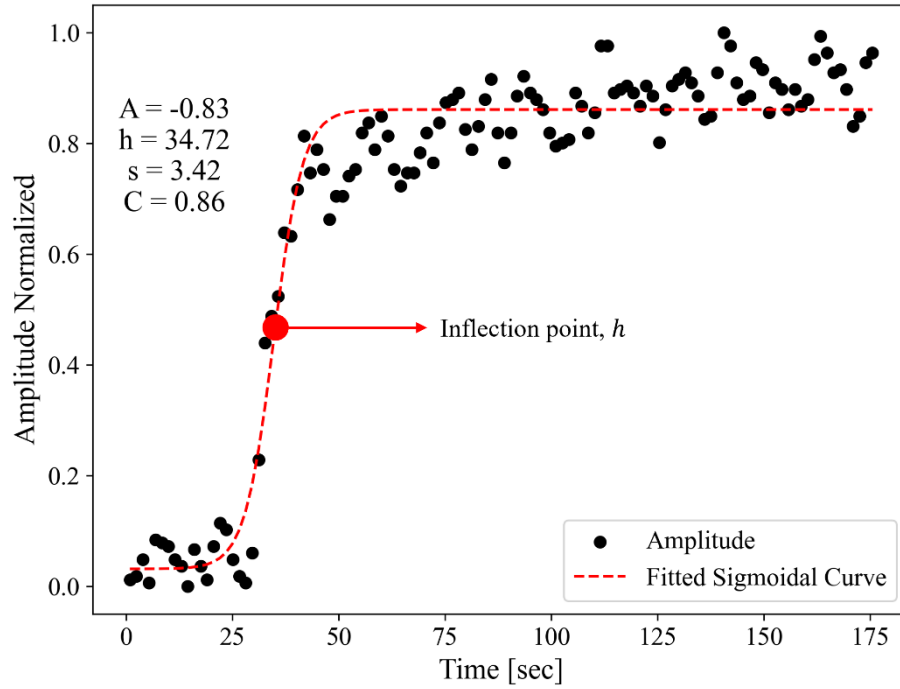


Figure 20: Normalized torque amplitudes with sigmoid fit

Once the curve fits, the following steps determine the output properties of the ink. First, the cure time is found by subtracting 30 seconds (off time) from the inflection point, h . This point is halfway up the curing curve and represents the time of cure of an ink. This point was chosen as it is easy to extract from the data, and it typically represents when the ink has surpassed the gel point and is forming a solid. Second, the maximum temperature is found by simply finding $\max(T)$, where T is temperature array recorded by the thermal camera. Lastly, the realizability is found with four conditions: 1) the $t_{\max(T)}$ must be less than 120 seconds, 2) s must not be equal to 0, 3) a curve fit is possible from the data, meaning, the parameters can be determined, and 4) the h value must be between 30 and 120 seconds. The first condition is necessary because some inks take longer than two minutes to cure. If the time at the maximum temperature is greater than 120 seconds, then the ink was still reacting when the test ended. This is not desirable for DLP printing and

will be difficult to process in any form, deeming the ink unrealizable. The second and third conditions go hand in hand as they relate to the curve being fitted properly. The s value being equal to zero means that the sigmoid has become a right angle, which is not physically possible. And if a curve fit is not possible, then the data does not lend itself to a sigmoid the ink most likely did not cure in time, if at all. Lastly, the final condition makes sure the inflection point is physically reasonable as anything outside of the curing time range is not possible. Each property and condition for realizability for the ongoing example data is depicted in Figure 21.

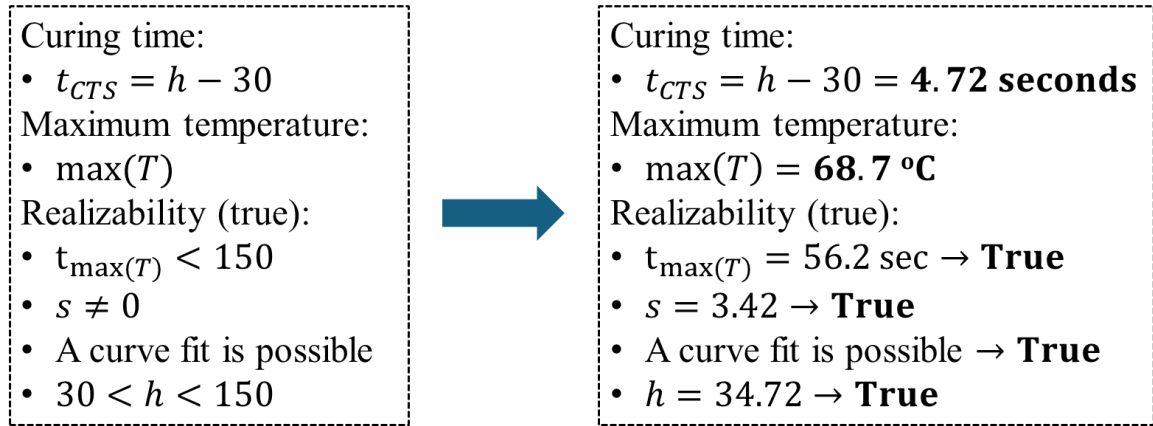


Figure 21: Three primary properties extracted from example in previous Figures 17-20

Considering both the temperature and torque data allows for a complete picture of the curing kinetics of the system regarding viability for DLP printing. From the temperature data, the maximum cure temperature, average cure temperature, and the Boolean status of curing ink are output. From the torque data, the cure time and Boolean status of the curing ink are output. From there, the realizability of the ink can be determined.

2.5.3 DLP Relationship

The cure time of the ink from the CTS can provide information about the DLP curing time. The main differences between the two systems boils down to Beer-Lambert Law.

$$\frac{dI}{dx} = -\mu I$$

where I is the light intensity, μ is the “optical density” or absorption of the body, and x is the distance. As an approximation, we assume μ is a constant across the thickness. When integrated, it yields the equation below.

$$I = e^{-\mu x}$$

From here, each system, DLP and CTS, must abide by the law, leading to

$$I_{DLP} = e^{-\mu h_p} \text{ and } I_{CTS} = e^{-\mu h_c}$$

The typical layer height, h_p , for DLP is $50\mu\text{m}$, whereas the layer height for the ink in the glass container, h_c , is approximately 0.32cm , orders of magnitude difference. The absorption, μ , will be the same for identical inks in each system as it depends on the composition. The absorption may change between compositions, but the difference can be ignored for this rough estimation. Using the equations for each system, the degree of conversion, or degree of cure (DoC), can be assumed proportional to the light dose, It . The h_p is much smaller than h_c and can be assumed as zero.

$$\frac{I_{CTS}}{I_{DLP}} = \frac{e^{-\mu h_c}}{e^{-\mu h_p}} = e^{-\mu(h_c - h_p)} \approx e^{-\mu h_c}$$

To reach the same DoC for both systems, the ratio between them can be found.

$$I_{CTS}t_{CTS} = I_{DLP}t_{DLP} \rightarrow \frac{t_{DLP}}{t_{CTS}} = \frac{I_{CTS}}{I_{DLP}}$$

A rough estimation of the cure time is shown below.

$$t_{DLP} = t_{CTS} \frac{I_{CTS}}{I_{DLP}}$$

To evaluate the values for the light intensities, a light meter was used for both systems. Figure 23 shows the experimental setup for both the CTS and DLP systems. The light meter was placed on the exit of the light from the ink to find the resulting light intensity after decay, I . The DLP system has little to no decay because the layer is so thin. Therefore, the entrance and exit light intensities can be assumed to be the same, which is a value of 10.11 mW/cm². For the CTS system, the layer height is very thick and the light decays significantly. The resultant intensity is measured as 1.03 mW/cm². The quotient of these two numbers yields the $\frac{1}{10}$ fraction for the comparison between the cure times of each system, i.e.,

$$t_{DLP} \approx \frac{t_{CTS}}{10}$$

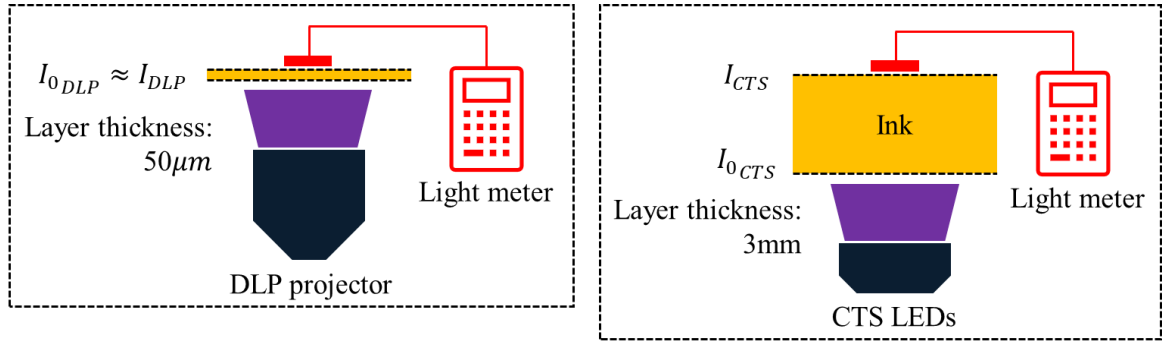


Figure 23: Experimental setup for the determination of light intensities between the DLP and CTS systems

Based on this assumption, experiments were performed to measure the curing time of the ink during DLP printing for comparison. The tests were performed by placing 50 μm shims between two glass slides with a layer of ink sandwiched in between. The two slides were clamped together, and UV light shined from a DLP projector into the ink. The cure time was measured by finding the time where the glass slides could no longer be separated,

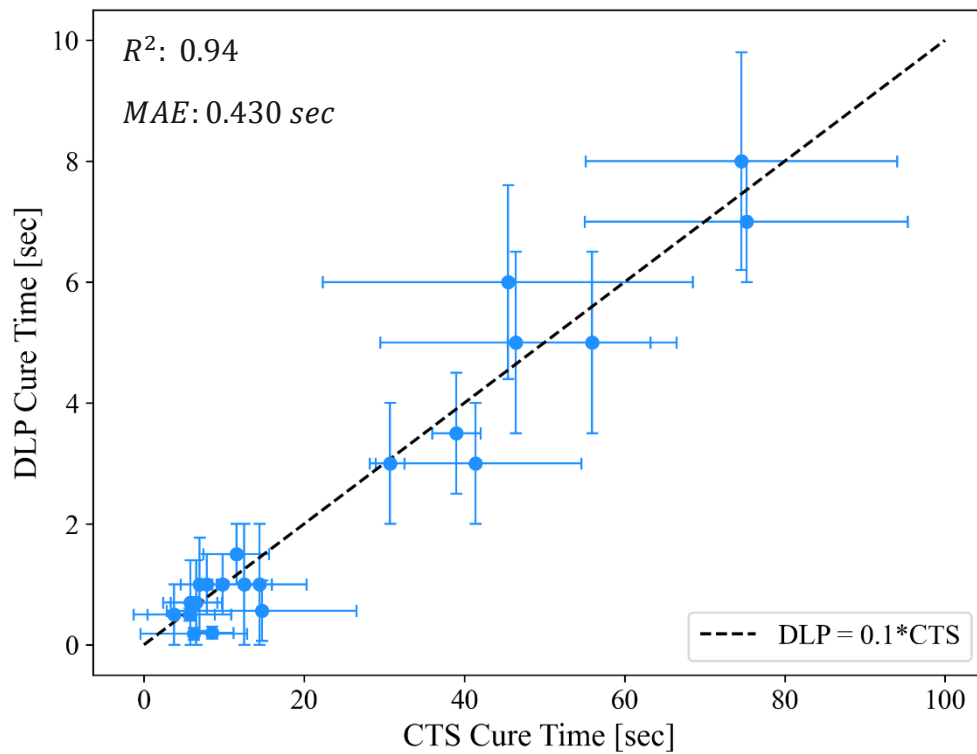


Figure 22: Experimental verification of CTS to DLP relationship

ensuring that the entire layer was cured. The plot of various inks and their resulting DLP and curing kinetics system cure times are shown in Figure 22. As can be seen, although there are large errors, the general trends follow the estimation of t_{DLP} being only 1/10 of t_{CTS} .

2.6 Automated Tensile Testing (ATT)

The uniaxial tensile testing system is the primary source of data from the samples. Each tensile test outputs the Young's modulus, ultimate tensile strength (UTS), and elongation at break, as well as the dimensions of the sample. The primary sensor for the tensile data is the Epson Force Guide mounted to the end of the robot arm. The forces, torques, and spatial positions in X, Y, and Z are tracked by the robot or force sensor to provide most of the necessary information to calculate mechanical properties. All tests are performed at a displacement rate of 1mm/s. The workflow of the tensile testing process is depicted in Figure 24. Each step in the process will be explained in the subsequent sections.

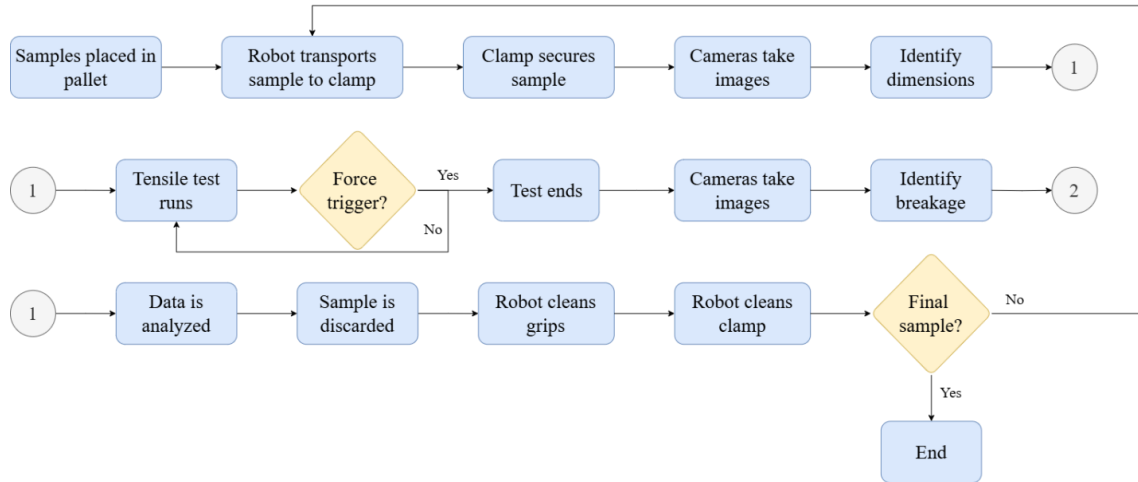


Figure 24: Automated tensile testing workflow

2.6.1 Automated clamps

In a uniaxial tensile test, there must be two clamps which hold either end of the sample during testing. Typically, these clamps are held together by a spring force and are manually opened and closed to hold a sample in place. Here, the clamps are automated and require no human intervention to secure a sample. The top clamp is the Schunk EGP 64-N-N-B (Schunk, Germany), which is an electric I/O parallel gripper with adjustable grip force between 50 and 300N and is mounted to the end of the robot. However, the grip force is set to 50N and left constant for all tests. The maximum stroke per jaw is 10mm and allowed a finger length of 80mm. The fingers used were custom made and machined from

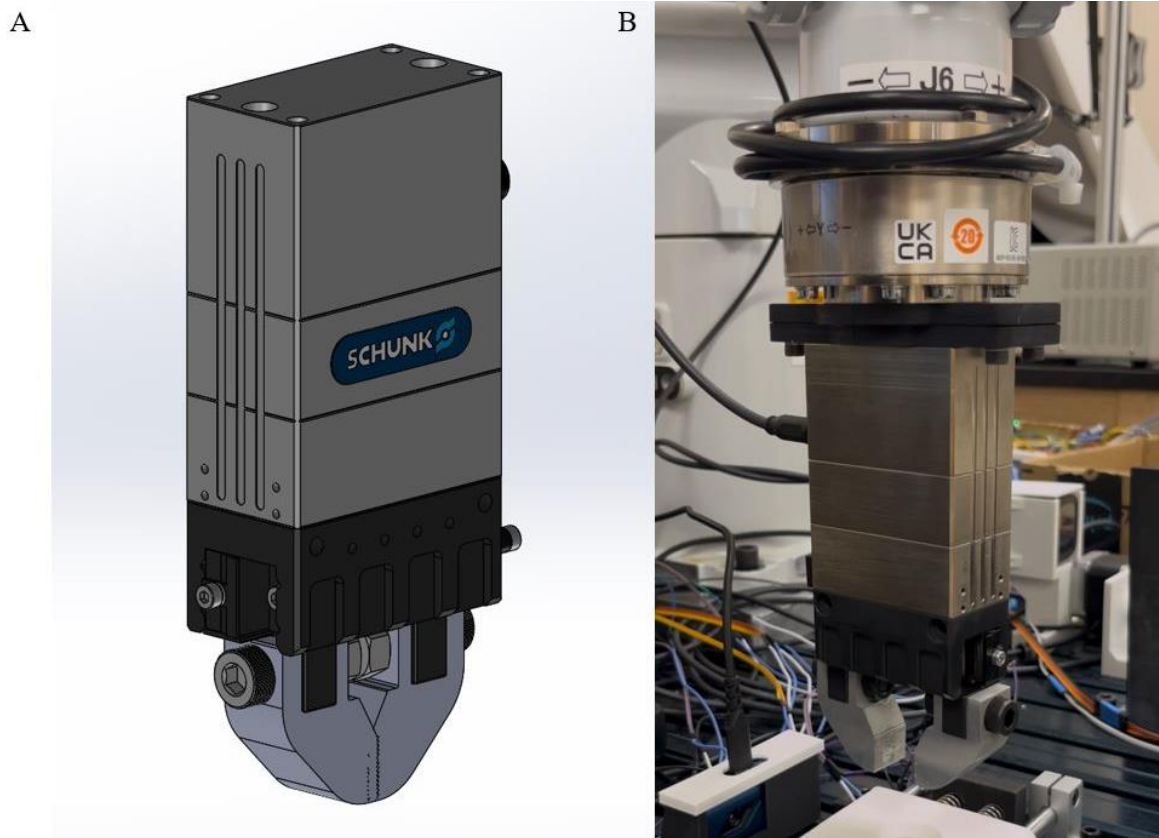


Figure 25: A) Schunk EGP 64-N-N-B electric gripper with Al fingers and B) gripper implementation with force sensor attached to robot wrist

Aluminum with small serrations to assist in gripping the stiffer samples to avoid any slippage. In addition, the gripper is powered by an AC servomotor that provided a continuous clamp force when in the “on” mode, ensuring the object being gripped is always subject to the 50N force. The gripper illustration is shown in Figure 25 with the real implementation and force sensor.

The other gripper is mounted to the table and is bespoke with a stepper motor, lead screw, linear motion stage, and removable grip surfaces. The design and implementation of the gripper are depicted in Figure 26. The gripper contains two main pieces that are mounted to the table. One piece holds the stepper motor and the guide rod, the other holds both the lead screw and guide rod as well as being the other clamp surface. The third piece is mounted to the lead screw and is the motion stage that actively clamps against the sample. The motion stage and the other clamp surface both have mounting holes to secure small aluminum grips with similar serrations to the robot gripper fingers. Again, the purpose is to grasp the samples securely so there is no slippage during tensile testing. In addition, there is an incremental rotary encoder attached to the other end of the lead screw to enable closed loop motion control. As samples can vary in hardness, the closed location of the motion stage clamp will not always be constant. Therefore, the position is actively tracked with the encoder to ensure it tightly secures the sample and opens the proper distance each time. Specifically, each time the system starts, the motion stage clamp is pressed against the other clamp surface, and the encoder is reset to 0. Then the motion stage

opens to a specific distance so the sample can be placed by the robot in between the two grip surfaces. As the motion stage clamp closes on the sample, the software checks for the position of the encoder and when it reaches zero, or it idles between a certain range. For example, if the sample is very stiff, the clamp will not reach the zero position, but when it securely presses the sample, the encoder position will remain somewhat constant for a certain amount of time, triggering the motor to stop as the end position was reached.

The limitation of the system is there is no constant force being output on the sample by the clamp, unlike the robot gripper. The strength of the clamp is limited by holding torque of the stepper motor. However, the limiting factor of the overall tensile system is the force limit of the force sensor on the robot, which is 250N. Because the gripper also adds weight ($\sim 50\text{N}$) to the force sensor, no samples which can handle a force of more than 200N are used, to ensure the force sensor is not damaged during testing. The current supplied to the stepper motor in the lower gripper is set to maintain a grip force lower than 200N. Therefore, if a sample is tested that exceeds the force limit, the sample will slip from

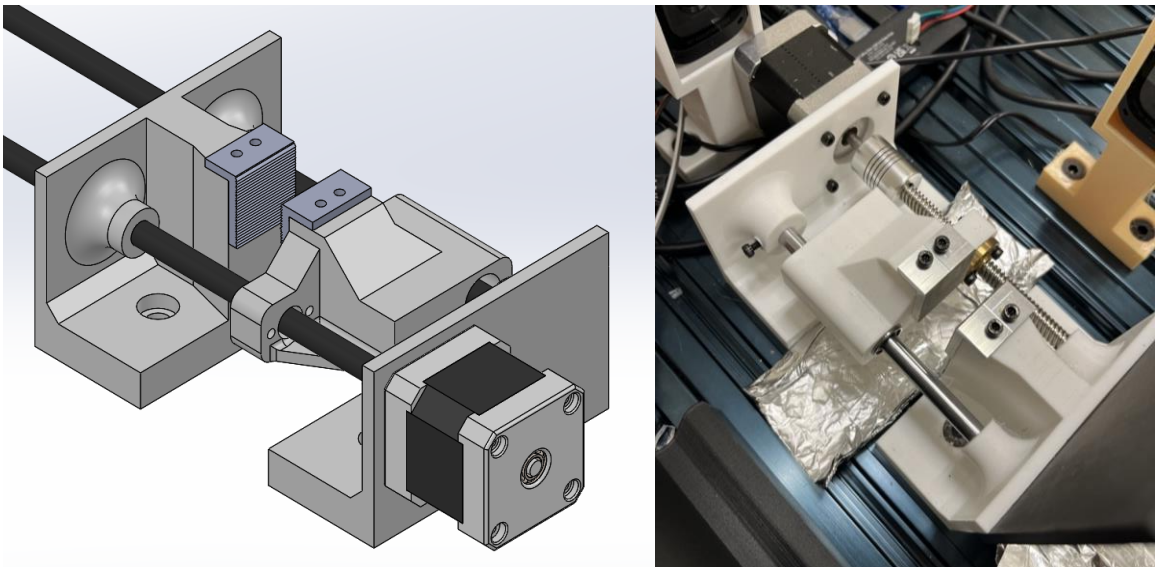


Figure 26: Automated clamp model (left) and real implementation (right)

the lower clamp and not damage the force sensor. The code contains logic that marks a sample when the force exceeds 150N to indicate that slippage was possible.

2.6.2 *Dimension detection*

The second most critical component for tensile data collection is the sample dimension detection. Each sample must have associated width and thickness measurements for calculating the cross-sectional area. For engineering stress and strain, the original width and thickness are used for calculating all the stresses. For true stress and strain, the real width and thickness are measured throughout the test to find how the area changes.

$$\sigma_{Eng} = \frac{F}{A_0}, \quad \varepsilon_{Eng} = \frac{\Delta L}{L_0}$$

$$\sigma_{True} = \frac{F}{A_i}, \quad \varepsilon_{True} = \ln\left(\frac{L_i}{L_0}\right)$$

For any sample, the initial width and thickness measurement is taken with the same method. There are two cameras mounted to face the front and side of the sample with black backdrops for each camera as illustrated in Figure 27. The cameras are the Anker PowerConf C200 2K webcams (Anker Innovations, Changsha, Hunan, China). They were chosen because of the USB interface, high resolution, and the ability to take images through the OpenCV Python package [22]. In addition, cameras are a non-contact method of measuring dimensions and will not have issues with samples of varying stiffness.

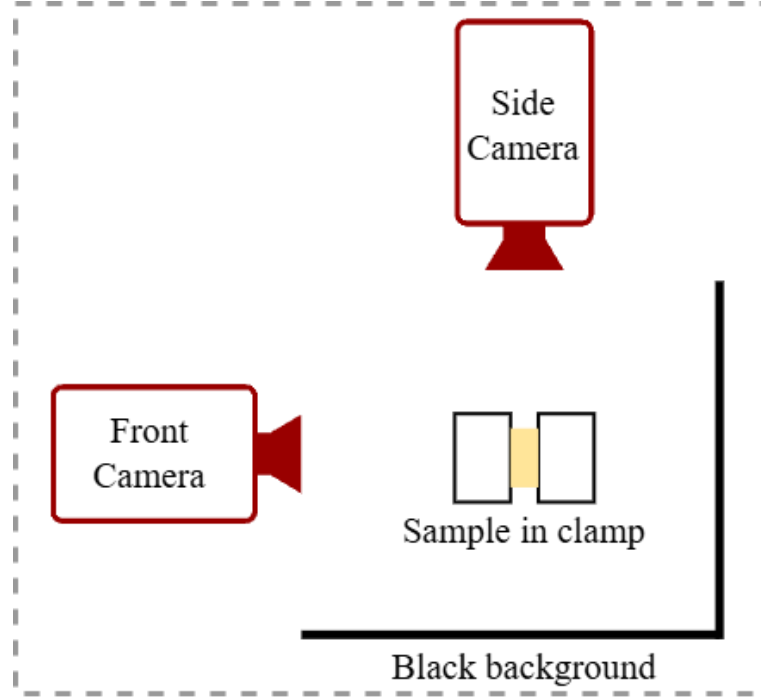


Figure 27: Top view of the dimension detection apparatus

Once the sample is placed in the automated clamps, both cameras capture an image of the sample from their respective angles. Each image is checked for two cases: 1) reasonableness of the dimension and 2) blurriness, which are not mutually exclusive. For the first case, the expected nominal dimensions of the samples for width and thickness are 4 mm and 2.5 mm, respectively. These are the dimensions of the mold in which the samples are cured. Therefore, there is some expected variation from this value, but it is constrained to 0.8 mm from the nominal. If a dimension outside of this range is detected, the image will be retaken. To detect blurriness, the Tenengrad focus measure is used as it was found to be the most consistent, which is the mean gradient magnitude of an image [23].

$$\text{Tenengrad} = \sum_{x=2}^{M-1} \sum_{y=2}^{N-1} [\nabla S(x, y)]^2 \text{ for } \nabla S(x, y) > T$$

$$\nabla S(x, y) = [\nabla S_x(x, y)^2 + \nabla S_y(x, y)^2]^{\frac{1}{2}}$$

If the value meets an experimentally determined threshold, then the image is determined to be in focus, if not, another image is taken. In an unlikely case, if no usable images or dimensions are captured after a series of ten attempts, then the nominal value is assumed.

After the image is determined to be acceptable, the processing ensues to extract the dimension. The primary steps with example images are shown in Figure 28. The image is first cropped to only include the sample. The cameras and bottom gripping apparatus are fixed in place, so the image is cropped to include nothing but the sample and the black backdrop to provide contrast. Then, Canny edge detection is used to find the edges with high sensitivity, meaning many edges are captured and the resulting image is very noisy. To remove the noise, Hough line method is used to extract vertical edges based on a threshold length as it is expected that the tensile bar samples will have straight edges. From the remaining vertical lines, the two outermost lines are chosen as being the outermost edges of the sample. Then the average distance between these two lines at a series of points is calculated to find the dimension of the sample. The same method is used for both side and front dimension detection algorithms.



Figure 28: A) raw image of a sample before testing from the front camera. B) Canny edges of the sample filtered with Hough lines. C) Detected outer dimensions and measurement points overlaid on the sample

The dimension detection was verified by examining the distributions of dimensions of samples measured manually with calipers and with the cameras. Further, the method was to take the dimensions of the sample with calipers before placing them in the ATT system, then record both dimensions. The dimensions were first examined separately, into width and thickness, to verify the accuracy of each camera. Initially, there was a slight bias in the cameras dimensions which were corrected, and the distributions were compared again to show similar central tendencies. Next, the cross product of the widths and thickness is taken for both the manual and camera dimensions. The resulting distribution is all the possible cross-sectional area measurements for the samples, which are shown in Figure 29 for both methods. The area is the value being used during the data analysis and encompasses both the width and thickness, making it important to verify.

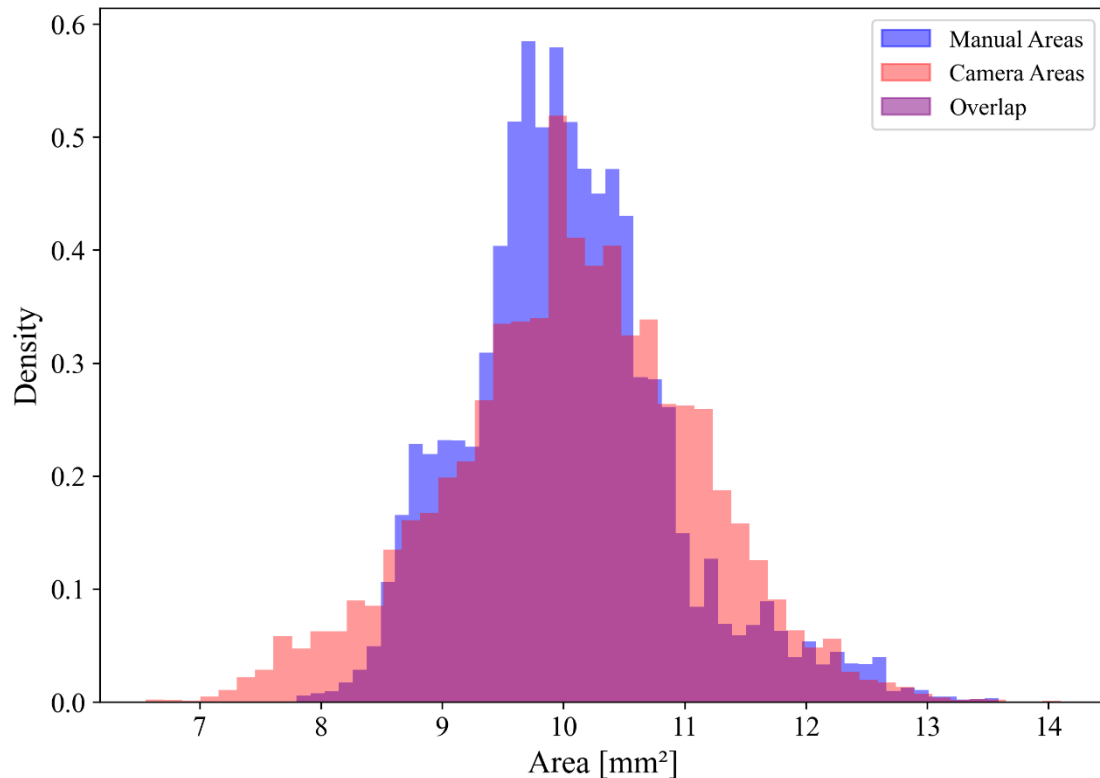


Figure 29: Cross product of the width and thickness dimensions measured by the cameras and calipers

The distribution metrics are shown in Table 1 below. The results indicate that the distributions are similar, with different shapes. The means are very similar, which demonstrates high accuracy of the system, and the standard deviations are similar, with the cameras being slightly higher. The skewness is very different where the camera dimensions are less skewed, and the manual measurements are right skewed. The Wasserstein distance was also found to demonstrate the similarities of the two distributions by quantifying the work necessary to transfer one distribution onto the other.

$$W_1(M, C) = \frac{1}{n} \sum_{i=1}^n |x_i^M - x_i^C|$$

Table 1: Comparison of dimension distribution metrics

Method	Mean (mm ²)	STD (mm ²)	CV (%)	Skewness	Wasserstein Distance (mm ²)
Manual	10.045	0.851	8.48	0.647	0.193
Cameras	10.075	1.034	10.263	-0.111	

The true stress and strain are found with the same edge detection method without the dimension limit condition, but for every frame of a video taken by each camera over the duration of a test. The dimension is expected to decrease as the sample is being stretched, therefore, the dimension limit condition cannot be used as the limits will change depending on how much the sample stretches. Figure 30 below shows an example of a sample that stretched and the resulting detected dimensions throughout the test.

There is no speckle pattern or complex digital image correlation method being used to find the dimension change; therefore, noise and potential inaccuracies are expected. The true stress and strain calculation serves as an addition to the engineering stress and strain.

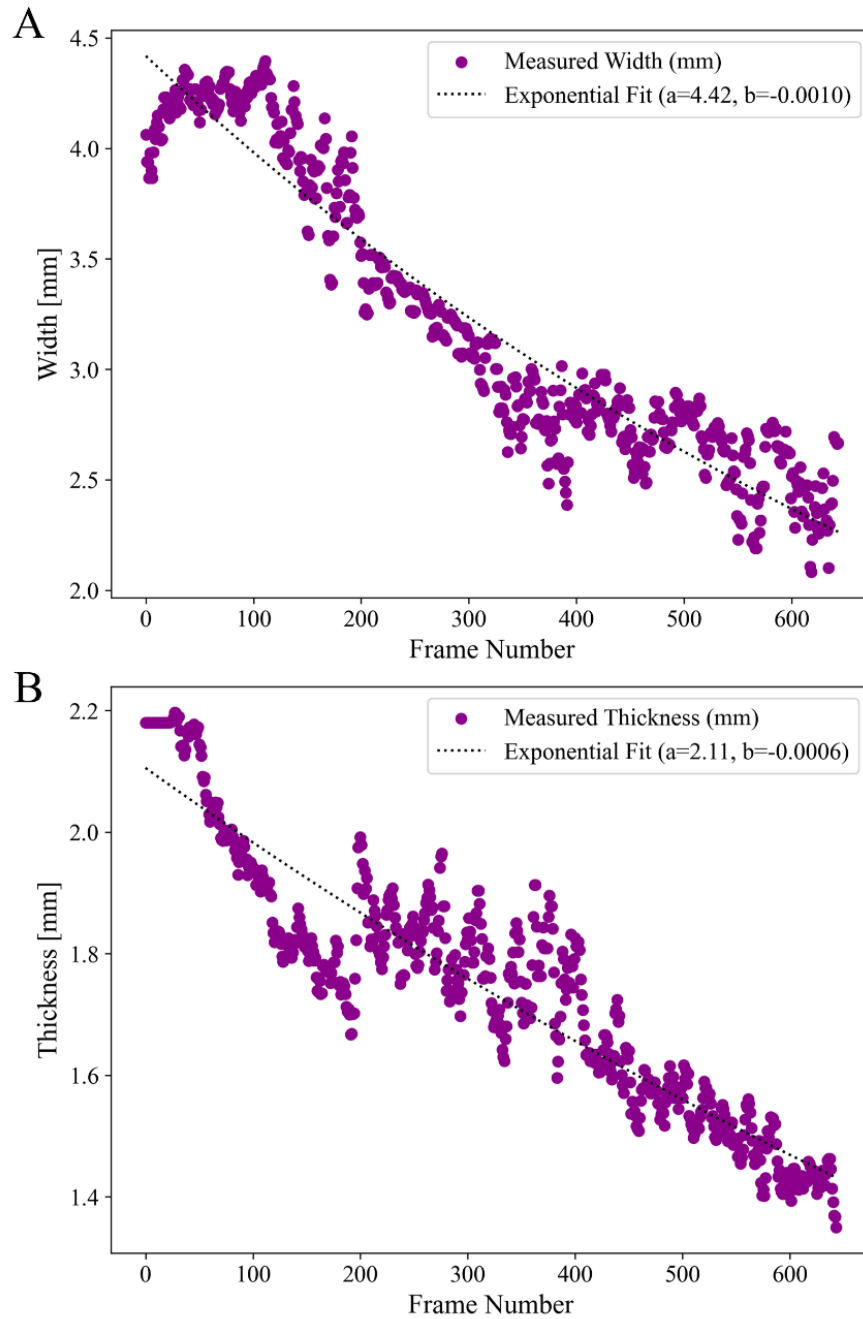


Figure 30: Edge detection for each frame of a video of a tensile bar test. A) front camera dimensions and B) side camera dimensions

2.6.3 Anomaly detection

Inspection of samples is typically performed by a researcher before and after testing. The objective is to find defects in the samples prior to testing or to observe where the sample broke after testing. Further, for “dog bone” style tensile bar samples, the ideal breakage point is in the gauge region where the sample width decreases relative to the two ends. Samples that break in the grippers may not provide accurate data as the sample may have had a defect, or in the case of soft samples, potentially was cut by the gripper. To automate this process, a model was implemented to perform anomaly detection using images of the samples after testing. Specifically, two convolutional neural networks (CNNs) were implemented for each camera which compared predictions. After a tensile test is completed, an image from the front and side camera, respectively, is taken of the bottom half of the sample in the lower grip. The image is passed through the model to output the class corresponding to the region of breakage. The four classes are: gauge region break, high break, low break, and stretched, the last class being that the sample did not break and was fully stretched. Figure 31 shows the process model for the system.

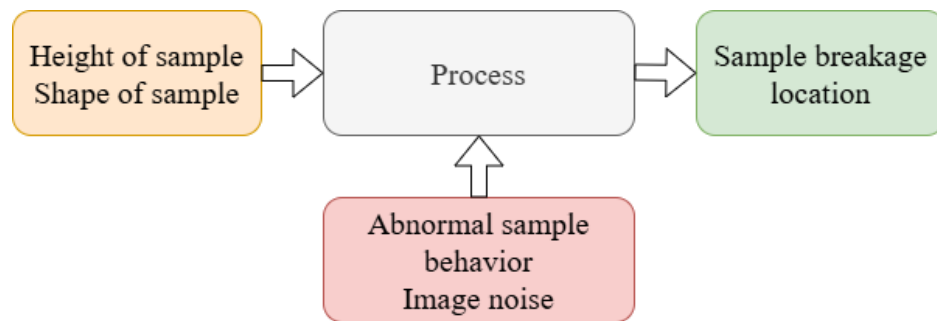


Figure 31: Post-test anomaly detection process model

Since the outputs are classes, the models are inherently classification models. The dataset consists of 660 images between the two cameras. The class distributions are as

follows: gauge break: 29.2%, high break: 28.0%, low break: 30.1%, and stretched: 12.7%. Initially, images were collected manually to build a dataset with approximately 40 images per class. Then the subsequent images were taken from real tensile tests, and the class distributions are indicative of which classes are seen the most. Therefore, the model will be well exposed to the important classes as the sampling is based on real data.

The images are initially preprocessed to make the classes more distinct and allow for better classification. Cropping is applied to remain consistent with all the images, and they are reduced to only the top portion of the bottom grip and the sample. In addition, the images are converted to grayscale to reduce the variations if samples have different colors, which is unlikely, but increases the generalizability of the model.

Each model is trained separately as the side images and front images are different. The training loss for both models is shown in Figure 32 over 35 epochs. This value was determined to be the most effective as any more epochs would start to cause overfitting. In

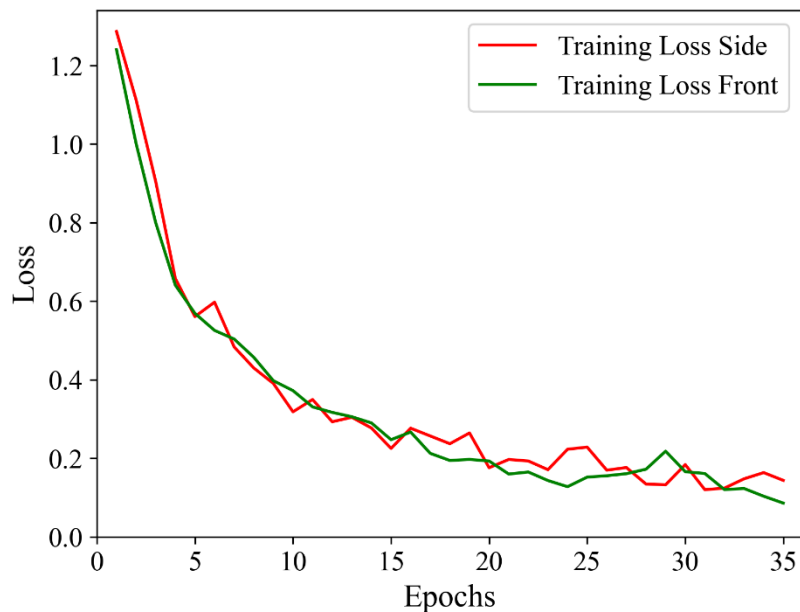


Figure 32: Training loss of both models

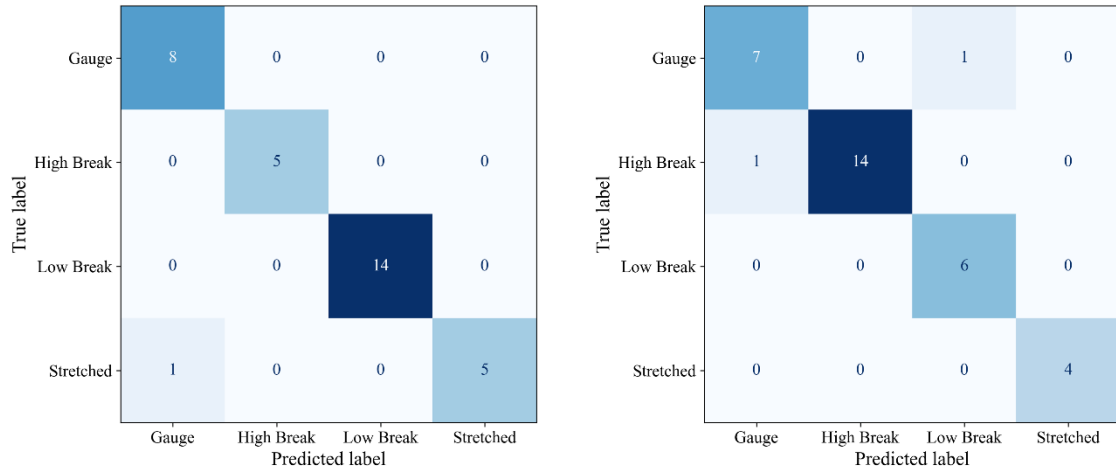


Figure 33: Confusion matrices for the validation set of model training for both the front (right) and side (left) models

addition, the resulting confusion matrices from this testing and validation loop are shown in Figure 33. The front model can discern the four classes consistently while the side model has some confusion, which is not entirely visible from the confusion matrix alone. The primary issue with the side model compared to the front model is the features visible. For example, the front model can see the curvature of the “dog bone” and learn which features of the sample correspond to which regions of breakage. Whereas the side model can only see the constant thickness of the sample and must use height alone to learn.

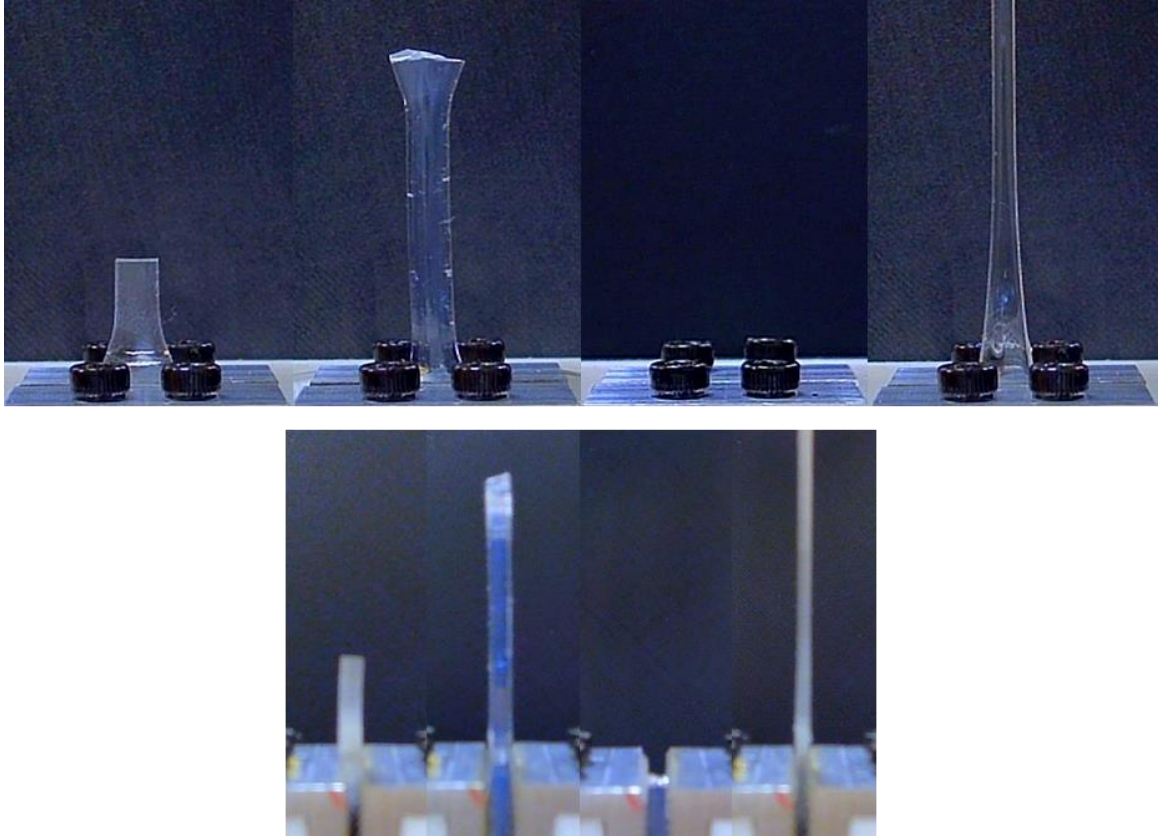


Figure 34: The validation images showing clear examples of each class for the front (top) and side (bottom) perspectives

The results of the training and validation indicate a strong classification model. The training loss converges slowly, demonstrating parameter tuning within both CNNs. The confusion matrices show that the model can correctly predict most cases, albeit for a couple of samples. For the side model, as mentioned, there is some understandable issue with samples in borderline cases. The incorrectly guessed gauge region sample being a high break is likely because the sample broke high within the gauge region. Meaning, the sample broke at the point where the curvature begins. From the front perspective, the model would be able to discern the shape of the dog bone and correctly classify it, but the side perspective cannot. The same argument fits for the misjudged gauge region sample as well.

To account for the discrepancies between the two perspectives of the cameras, an ensemble method is employed. After a tensile test is conducted, both cameras capture images and make predictions on the resulting class. The predictions are then compared by examining the probabilities of each class. The average probabilities are calculated, and the corresponding highest probability class is chosen as the final prediction. An additional validation is performed to demonstrate the effectiveness of the ensemble method and how it alleviates the side model issues. Clear cases of each class that have never been seen by the model were chosen for the validation and are shown in Figure 34. The predictions and probabilities for each prediction are shown in Table 2. The front model is very confident in its predictions as shown by the high probabilities. Additionally, the side model is confident in all predictions besides the final class. The height of the sample is again the culprit in this situation because from the side, a high break sample looks very similar to a stretched sample. In addition, there are many more high break samples than stretched samples in the dataset as that does not occur often. Therefore, the model tends to bias towards the more common class when it is unsure.

Table 2: Comparison of predicted classes and probabilities

Front Prediction	Probability	Side Prediction	Probability	Final Prediction	Ground Truth
0	84.8%	0	93.9%	0	0
1	97.5%	1	91.7%	1	1
2	99.3%	2	95.9%	2	2
3	94.8%	1	32.8%	3	3

2.6.4 *Tensile data analysis*

Typically, tensile data is analyzed manually and observing the curve can provide information about the behavior of the sample. For use in the SDL, the data analysis must be completely autonomous and reliably provide resulting mechanical properties regardless of the behavior. The primary properties of interest are the: Young's modulus, ultimate tensile strength, and elongation at break. These three properties help to determine the strength, stiffness, and stretchability of the material, which are relevant for polymer samples. Each time a tensile test is conducted, the resulting force and displacement data are saved and used in conjunction with the dimensions found by the cameras to calculate engineering stress and strain. The true stress and strain are calculated using the dimension changes from the videos of the tensile bars.

To find the Young's modulus, the first and second derivatives of the stress/strain curve are calculated. It is assumed that a typical elastic region for the Young's modulus will be a straight line with a relatively constant slope, making the second derivative in that region theoretically zero. Then, the cumulative change of the second derivative is used to find when it exits a certain threshold, which is set as the average of the first two percent of the data. Once the cumulative change exits that threshold, the elastic region is assumed to have ended. The slope of that region is then calculated to find the modulus. This method can reliably produce a modulus value and performs especially well for samples with clear elastic regions. However, many polymer samples, such as elastomers, will not show clear elastic regions. The modulus is calculated in the same manner, but it will tend to use less points as the threshold region is exited quickly.

The ultimate tensile strength is calculated as the maximum value of stress during the test and can be executed easily and consistently. The elongation at break is similarly simple as it is merely the strain percentage value when the sample is broken. The robot force sensor has a trigger that will stop the tensile test and data collection when the force exits the bounds of -0.5N and 200N. The lower bound is to detect a breakage in the sample. When a tensile test ends, there is a reactionary force exerted on the gripper in the opposite direction of the test. Therefore, there will likely be a negative force applied which can be used to detect the end of a test. However, in some cases, the sample is very soft and will not cause enough reaction to exit the bound and cause the force trigger to occur. In this case, the robot will continue performing the tensile test and collecting data despite no sample being present. The remaining data must be removed as it can contaminate the

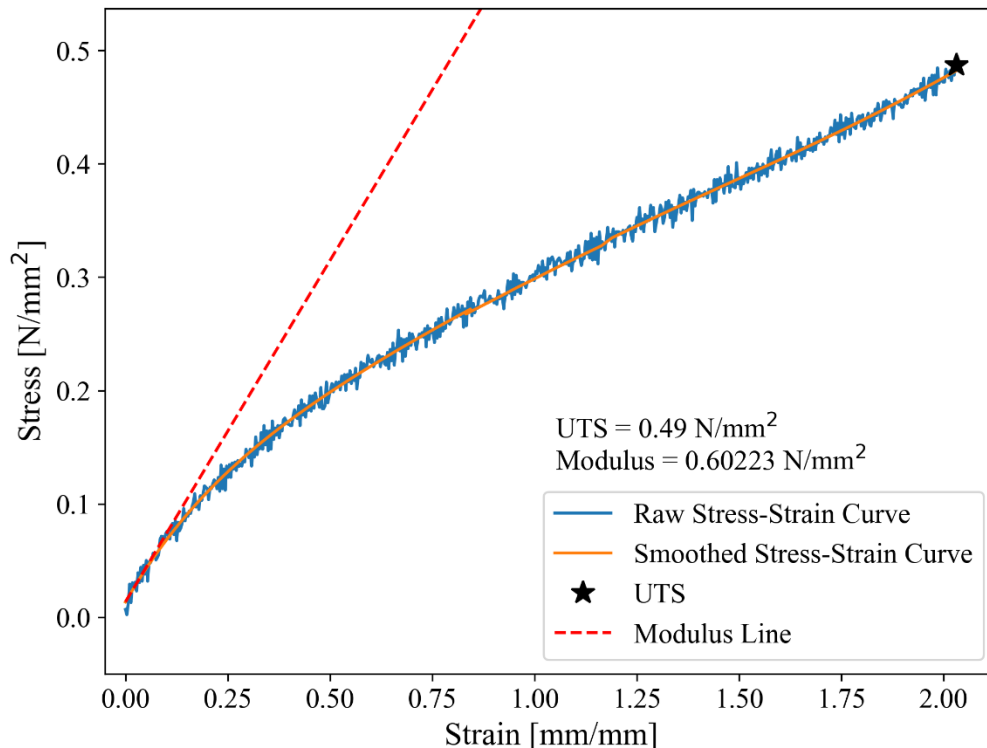


Figure 35: Example tensile curve with autonomous analysis performed

elongation measurement. In the data, the sample breakage is clear as there is a sudden drop in stress. In terms of the derivative, there is a large spike at this point, which can be detected easily. When this drop occurs, the code cuts off the remaining points and keeps the final point as the breakage point. Figure 35 is an example tensile curve that shows each property and how it was detected. In addition, being that the example was a soft sample, it shows a nondistinctive modulus region and how the modulus fit was determined.

The true stress and strain can be found by using the detected dimensions from the tensile test video for each camera. The fitted curves can be applied to the raw tensile data to provide an estimate of the true stress and strain. The same data from the previous tensile data fire is used to show the change in behavior in Figure 36.

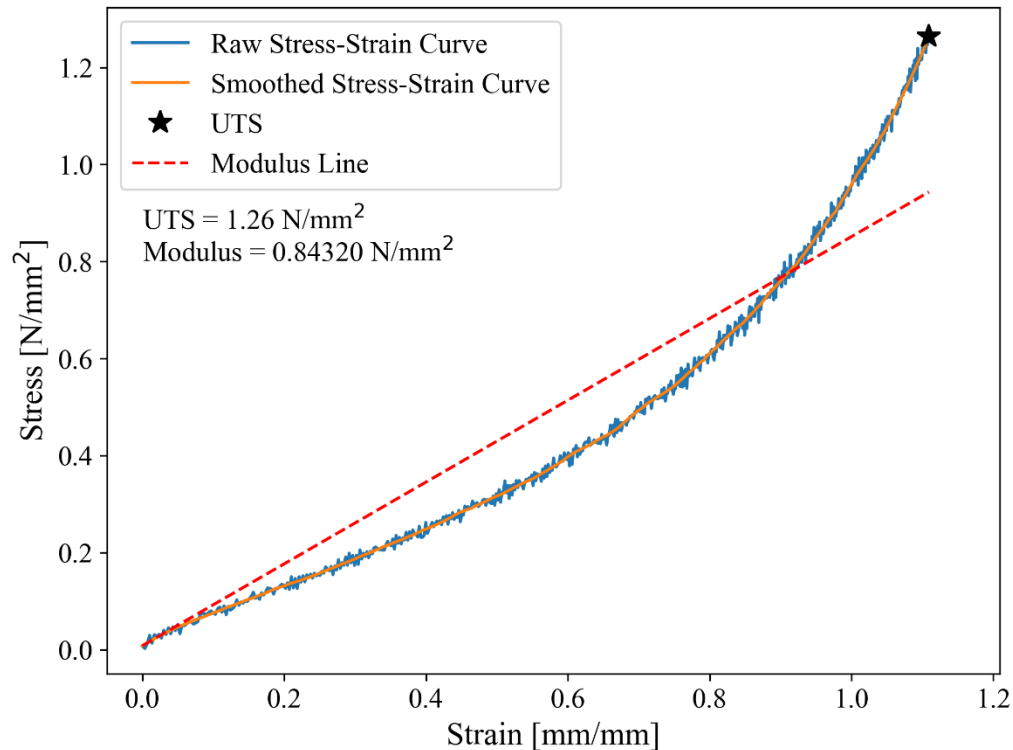


Figure 36: Example tensile data with true dimensions applied

Lastly, an important aspect to consider is the accuracy of the robotic tensile testing system. There are several factors that can lead to differences in data when comparing a commercial tensile testing machine and a robotic tensile machine. First, the displacement and deformation of the testing frame is key. Commercial tensile testing hardware is made with very stiff materials and designed to withstand the forces from a tensile test and not deform. Whereas the robot is also made to not deform, but the purpose is not typically for applications like tensile testing. The tensile machine has one degree of freedom, the vertical Z direction, whereas the robot has six DoF. To verify the accuracy of the robotic system, a test was conducted between the MTS Criterion C41 system (MTS, Eden Prairie, Minnesota, USA) and automated tensile testing system. Three samples were made and tested first on the robotic system, then let rest for several hours, then tested again on the MTS system. The samples were only stretched to 300% strain so they could be reused. There are no major distinctions between the two systems and their resulting properties which is illustrated in Figure 37. However, the “yield” point of the samples changed for the MTS system test. This is likely due to softening of the material after being tested

previously. The results also confirm that the robot is not greatly affected by the force undergone by the sample during the tensile testing.

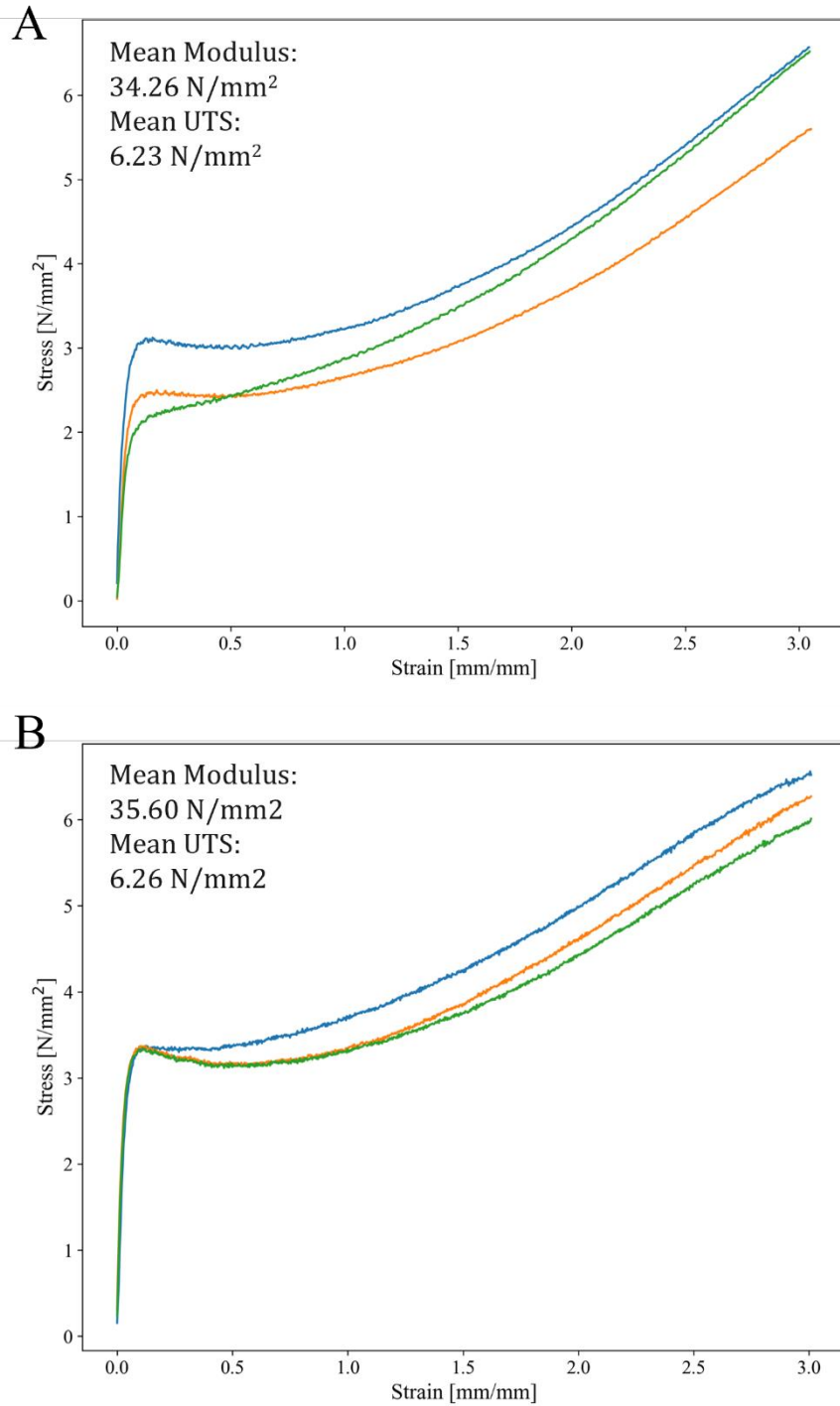


Figure 37: A) MTS system and B) robotic system tensile testing results

2.6.5 *Tensile bar processing*

The SDL described here is not level five as the entire system is not closed loop. There are automated workflows, but the system cannot be run completely without human intervention. Further, human intervention involves making the tensile bars for the tensile testing system. The method of making tensile bars involves first making silicon molds. The tensile bar geometry follows the American Society for Testing and Materials (ASTM) standard D638 Type V for polymer materials [24]. Then, the inks are dispensed with pipettes into the tensile bar molds and cured with a UV lamp for two minutes. It is desirable to have the inks cured for a long time to ensure they are not still reactive and have completely solidified. Once the curing is done, the inks are manually removed from the flexible silicon molds and prepared for testing. The automation of this process is discussed in the future work section.

2.7 **Summary**

In this chapter, the overall SDL system was introduced. There are three main components corresponding to the material processing and characterization of samples, which are the material composition system (MCS), curing time system (CTS), and the automated tensile testing system (ATT). These are integrated with the robotic hardware, microcontrollers, and the server that drives the system.

CHAPTER 3. OPTIMIZED DISCOVERY OF PHOTOPOLYMERS

The application of the SDL will be discussed in this chapter. The characterization systems are used to test and gather data on various compositions of photopolymers to discover new materials and establish relationships between components.

3.1 Integration with Active Learning

There was an active collaboration with Mr. Ayush Jain in Dr. Rampi Ramprasad's group during the later duration of the project. Further, Mr. Jain developed the decision-making algorithm and database based on a previous collaboration with the research group. To integrate the SDL processing and characterization systems with the decision making, a GitHub repository was established. The tree structure of the repository is shown below in Figure 38. Each subsystem contains its respective files to run and communicate with the main PC that the SDL hardware connects to. The compositions are sent to the MCS system initially from the sever through the active learning communicator. Once the data is collected and ready to be sent back to the server, the data analysis is performed to ensure the data is in the correct format, then the active learning communicator is called again to process the data into the server and database.

The driver function runs the entire system from start to finish. As the SDL is not a completely autonomous level five lab, the driver does contain user inputs to run each

system. This is necessary as the systems must be prepared before testing, such as loading the inks/containers into the CTS system or placing the tensile bars for the ATT system. In addition, once the systems all run and the data is ready to be sent, there is a user input to officially push the data to the server. The intention here is that, if necessary, the user can review the data and make changes if there are any major issues during any of the subprocesses. In almost every case, except the first several trial iterations, the systems handle anomalies properly and the data is ready to be processed.

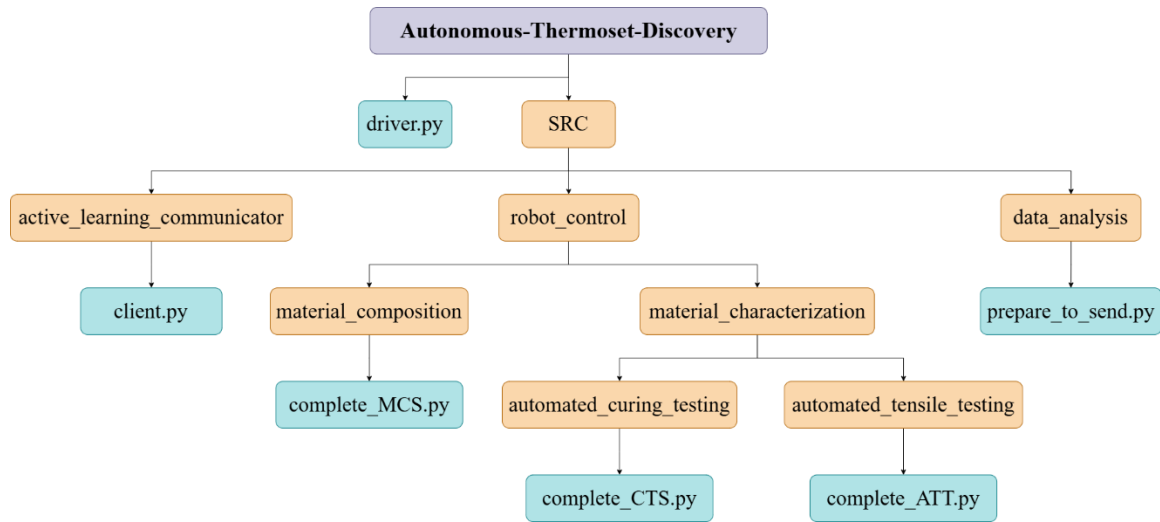


Figure 38: GitHub repository tree structure depicting the SDL software organization and integration with active learning

3.2 Model

The two goals of the SDL are to find ideal photopolymers for specific applications and discover new and unique photopolymers for DLP printing. Since the hardware and software are integrated and the SDL is functional, the outputs of the ATT and CTS system can be used to create a model to reach said goals. First, the forward process model is

depicted in Figure 39 with the monomer compositions as factors in their vector form, with each index corresponding to a specific monomer/crosslinker percent weight. The responses are the outputs from each characterization system, which are curing and mechanical properties of the compositions. The factors and responses were chosen because they represent the vital information during the photopolymer design process. These factors are necessary to create the proper formulation of the compositions. The responses characterize the material in a way that is usable for common goals. For example, if a DLP printable soft material is desired for stretchability, then the elongation response would be maximized, the modulus and UTS minimized, and the curing kinetics responses would be constrained to less than ten seconds curing time, less than 100C curing temperature, and realizable. A complete picture of the material is illustrated from these properties. The cofactors include the degree of cure of the samples and error from the autonomous systems. Breaking these down, the DoC of the samples can vary between samples in the same composition, which can be a potential issue for the tensile testing. Some errors can be introduced in the mechanical properties if one sample is cured less than another. A lower degree of cure implies the material is softer as fewer bonds have formed. Additionally, the error from the

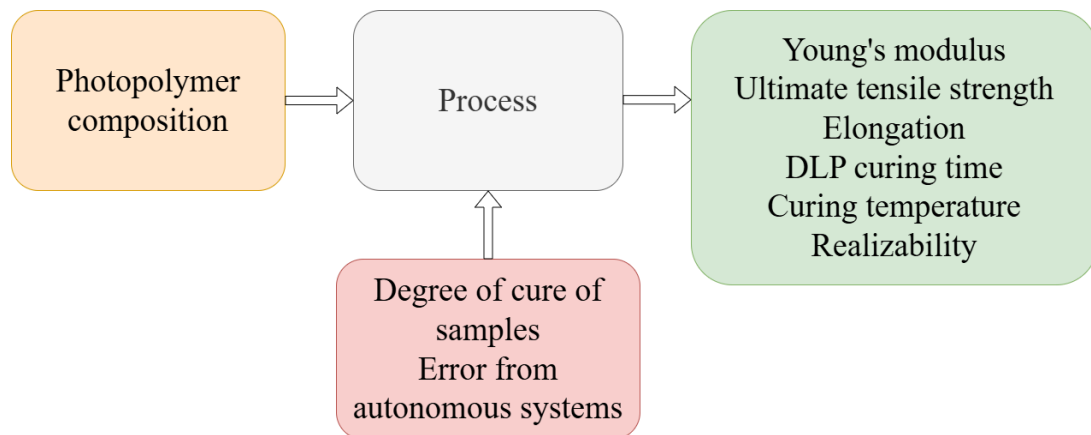


Figure 39: Forward process model of photopolymer discovery

autonomous systems is also present. Further, the characterization systems may have anomalies occur, such as the probe not being placed correctly in the CTS system, or the grip not closing correctly in the ATT system. There are safeguards in place to detect common anomalies and show them in the data, but the safeguards are not exhaustive. Since the systems are autonomous, rare anomalies that cause small changes may go unnoticed and infect the data for a certain sample or composition.

The forward process model fulfills the first goal of learning relationships between photopolymer components and their resulting properties. The second goal is to find ideal photopolymers for specific applications, which is fulfilled by the reverse process model shown in Figure 40. After enough data is collected from the SDL, the forward process model will be able to predict properties based on the composition. Then the desired response value can be used to find the photopolymer composition required to meet it. For example, if a modulus value of 100kPa is desired for a soft and stretchable material, the factors would represent the properties of this material. The model will output the proper composition to be tested.

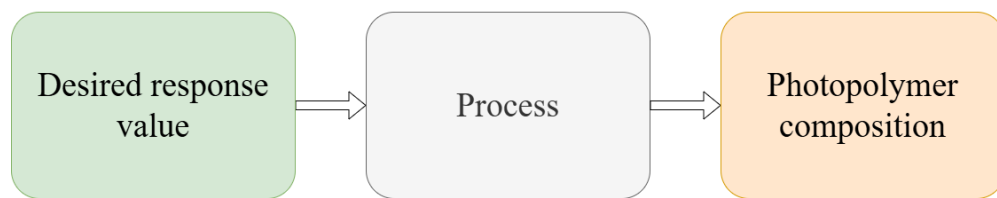


Figure 40: Reverse process model for finding ideal photopolymers for specific applications

3.2.1 Chemical Space

The chemical space was chosen to build upon the work by Jain and Armstrong et. al [1]. and encompass more materials and components commonly used in photopolymer research. Further, the additional components were chosen based on variety and property intuition. For example, the monomer Benzyl methacrylate (BMA) is known to cure very slowly during polymerization because of the heavy Benzyl ring. Inks with any amount of BMA will cure slowly even with crosslinker present. In addition, high amounts of BMA, any weight percent larger than 50, may not cure substantially within a two-minute period. Isobornyl acrylate (IBO) and BMA are also known to be stiffer components as adding them

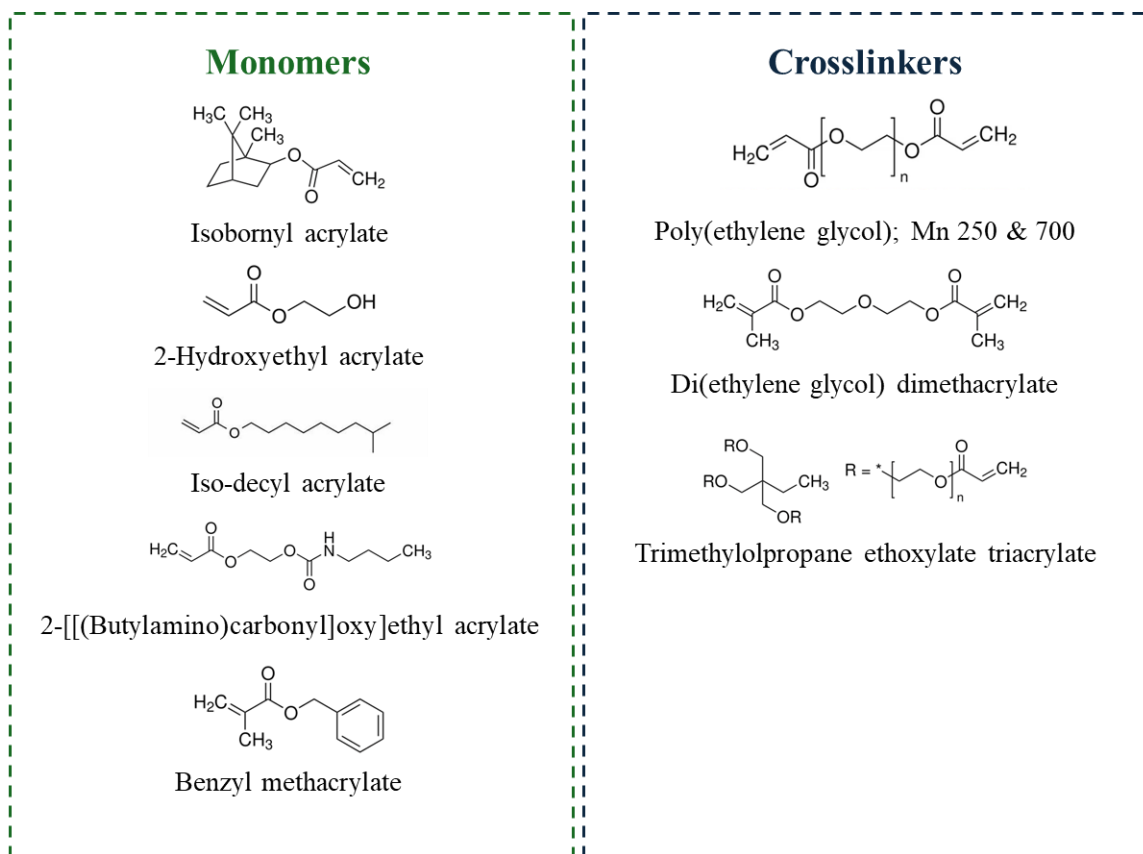


Figure 41: Respective names and structures for the monomers and crosslinkers chosen for the chemical space

to a mixture will typically cause the modulus to increase. The remaining monomers, 2-Hydroxyethyl acrylate (HEA), Iso-decyl acrylate (IDA), and 2-[[[(Butylamino)carbonyl]oxy]ethyl acrylate (BCO), are known to be softer or more stretchable and can create interesting properties when paired with the stiffer components. Lastly, the crosslinkers are chosen as they are all different in structure. Poly (ethylene glycol) diacrylate (PEGDA) is common and is used in many acrylate resin literatures. Both Mn 250 (PEG) and Mn 700 (PG7) versions of PEGDA are used here. Di(ethylene glycol) dimethacrylate (DEG) is a methacrylate, and Trimethylolpropane ethoxylate triacrylate (EPT) is a triacrylate, distinguishing them from PEGDA which is a diacrylate (Sigma-Aldrich, St. Louis, MO, USA). Each of the monomers and crosslinkers and their respective structure are shown in Figure 41. Moreover, every composition contained its respective weight ratios of monomers and crosslinker and 1 wt% photoinitiator Irgacure 819 (bis(2,4,6-trimethylbenzoyl)-phenylphosphineoxide) (Sigma-Aldrich).

Given all nine components, five monomers and four crosslinkers, the total number of combinations with no restrictions is practically infinite. Therefore, the constraints of each composition having two monomers and one crosslinker are introduced. Also, the weight percentage intervals are limited to 5% in weight. The total number of possible combinations in this space becomes 6840 photopolymer compositions. It would be expected that data from hundreds of compositions would be necessary to create a predictive model from this space, which would require nearly ten thousand manual tests if an SDL was not present.

3.3 Results

The SDL integrated with active learning facilitates efficient data collection for photopolymer compositions. At the time of writing, a total of 58 compositions were tested and the results added to the database to train the model. All compositions except two were selected by the active learning exploration algorithm. The two non-recommended compositions were added as they are common lab inks and fit the design space. The system is run constantly, and more data is added to the database daily.

3.3.1 Efficiency

One of the primary considerations for SDLs effectiveness is the ability to accelerate the research goals at hand. Further, two methods of quantifying the acceleration of the goals are by time saved and samples reduced, the latter implying money saved as well. The former will be focused on in this section as it can be easily shown. Several metrics are shown in Table 3 below describing the efficiency of the systems. It is important to note that the tasks performed are autonomous, and supervision is not required. Therefore, the time

Table 3: Efficiency metrics of the photopolymer SDL

Metric	Value	Material Composition System	Curing Time System	Automated Tensile Testing
Time per sample	t_s	15 min	4 min	5 min
Samples per composition	s_c	1	3	8
Samples per iteration	$s_i = s_c * 4$	4	12	32*
Time per iteration	$t_i = t_s * s_i$	60 min	48 min	160 min
Total iterations tested	i_T	14	14	14
Total samples tested	$s_T = i * s_i$	-	174	480*
Total run time	$t_T = i_T * t_i$	14 hrs	11 hrs	37 hrs

shown is not to be compared with manual times of a user performing these same tasks, as the researcher can be performing other tasks while these are occurring. The total amount of time to test the compositions was approximately three weeks.

The asterisk is present on the two metrics for the automated tensile testing because these values may not always be correct. For each iteration, not all the samples are guaranteed to be realizable, therefore, there may be less samples in each iteration. Also, since there have been 13 non-realizable samples present in the data so far, the total number of tensile bar samples tested is lower than the theoretical 480 value, at 376.

To evaluate the SDL further, the time metrics can be compared to the time to perform the same tasks manually as well as showing the true human time of the SDL. Human time is defined as the time where a researcher needs to be present and working. Since the time spent performing the tasks in each SDL subsystem is without human intervention, the run time of the SDL systems can be ignored. In Table 4, the relative improvement of the SDL is shown. The SDL frees hands-on time by 91.3%, which allows more than 6 hours of a workday to be allocated to other tasks. In addition, the time-consuming task of manual data analysis for both characterization systems is reduced by 100% because of the complete automation.

Table 4: SDL performance and automation metrics per iteration

Metric	Value	Photopolymer Processing	Curing Time Testing	Tensile Testing	Data Analysis	Total
SDL run time	t_{SDL}	60 min	48 min	160 min	0 min	268 min
Human time	t_{human}	5 min	10 min	20 min	0 min	35 min
Manual time	t_{manual}	40 min	60 min	180 min	120 min	400 min
Human time saved	$t_{manual} - t_{human}$	35 min	50 min	160 min	120 min	365 min
% Human time saved	$\frac{t_{manual} - t_{human}}{t_{manual}} * 100\%$	87.5%	83.3%	88.9%	100%	91.3%
Automation factor	$\frac{t_{manual}}{t_{human}}$	8x	6x	9x	∞	11.4x

3.3.2 Factors

The design space is being sampled based on uncertainty [25]. Further, the most uncertain samples are selected, then k-medoids clustering is used to get the centroids of the clusters. The centroids of the clusters are then recommended to be the compositions for the next iteration of testing. During this stage of data collection, there is no exploitation. Given the lack of previous data for this design space, the objective is to only explore the space

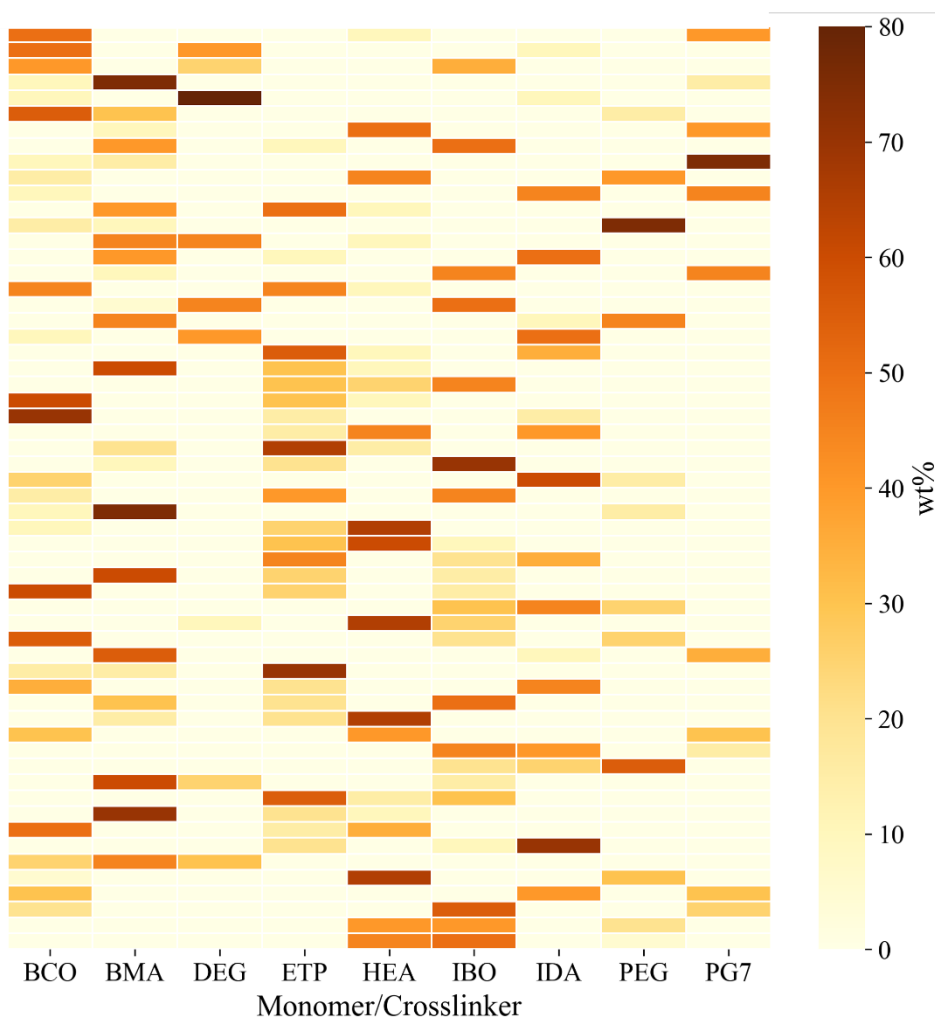


Figure 42: Factor heatmap for the first 58 compositions

and choosing samples based on the highest uncertainty is an effective and simple method to do so. The heatmap of factors is shown in Figure 42.

The heatmap shows the distribution of monomers to be consistent, with BCO occurring the most by only a couple of compositions. However, the crosslinkers are rather misrepresented as ETP is used in just under half of all the compositions. The reason is the uncertainty of the ETP samples. The samples are typically heavily crosslinked as ETP is a triacrylate, meaning, it has three reactive sites. Heavy crosslinking causes the material to be very stiff and brittle, and the tensile testing data is more inconsistent. Since the sampling is based on uncertainty, the algorithm was choosing ETP more often. An adjustment was made to the exploration algorithm to use the median values of uncertainties to create a baseline understanding of the overall uncertainty in the dataset. This allows the ETP uncertainty to not appear as an outlier that needs to be sampled further.

3.3.3 Responses

The responses of the factors in the model cannot be predicted intuitively. If they could, then a complex machine learning algorithm would not be necessary. As mentioned, the factors are chosen to sample many areas of the design space and not to bias the model. The objective is the same for the responses, although it is impossible to choose which responses to collect data on. The distributions of the responses can be analyzed to judge whether the sampling method is effective. Figure 43 shows the distributions for the six outputs of the model. The Young's modulus distribution shows a good mix of soft and stiff samples, with moduli ranging from kPa to GPa ranges. The UTS is heavily correlated with

the modulus and follows a similar distribution but with a smaller range. The elongation has the least desirable distribution of the responses. Most of the samples have a very low elongation and only one sample has surpassed 100% strain so far. The reason for this comes down to the crosslinker requirement in the design space. Further, the crosslinker causes a more solid network to be formed in the photopolymer, which in turn causes the stretchability to go down since the polymer chains cannot easily move. In addition, many soft materials would be expected to stretch and have high elongations, but the presence of crosslinker again limits their stretchability. Many of the resulting compositions are soft and

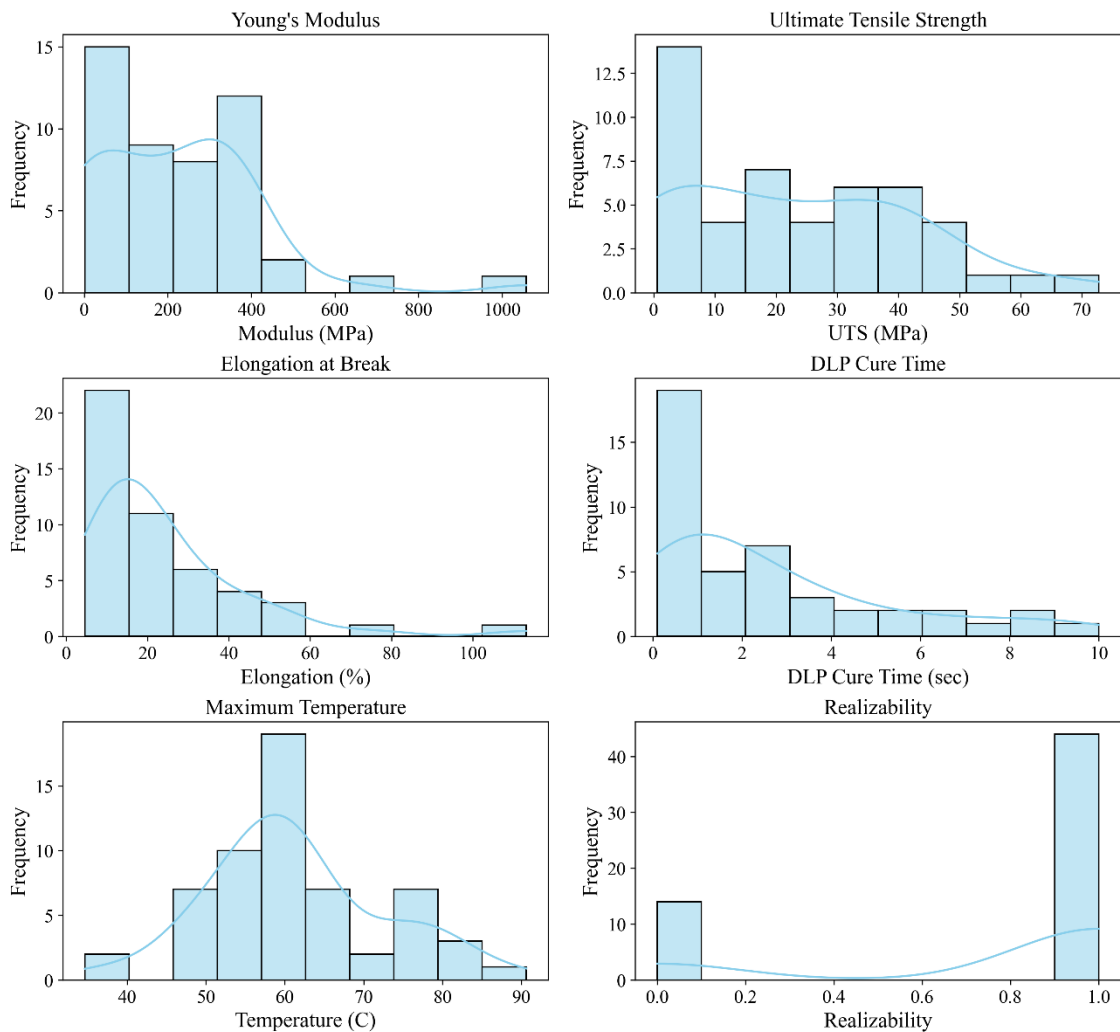


Figure 43: Response distributions for the first 58 compositions

non-stretchable, which are not ideal properties for most applications. The maximum temperature has the most normal distribution of the six and has seen a wide range of values. The cure time and realizability are related, and their distributions are the most surprising. Most of the compositions have a cure time below five seconds, which is excellent for DLP printing. In addition, most of the samples have been deemed realizable. Again, the crosslinker requirement is the culprit as the reaction will occur faster with crosslinker present in the mixture. While it may have some downsides for the elongation, it is helpful to ensure printability of the inks. As the model becomes a more effective predictor and learns the relationships between the factors and responses, the crosslinker requirement could be removed. For an initial sampling of data, it is expected that the values will not be dispersed greatly as the model is still exploring. As more data is collected, the range of distribution will expand because the more uncertain compositions will not be near the mean composition responses.

3.4 Summary

This chapter shows the addition of the decision-making aspect of an SDL to the existing processing and characterization systems from the last chapter. Compositions are selected by an active learning exploration algorithm and processed and tested by the photopolymer SDL. There is a heavy improvement in data collection efficiency and despite the design space being rather large. The data shows that the distributions are reasonable and the crosslinker requirement in the design space has a high effect on the responses.

CHAPTER 4. CONCLUSIONS AND FUTURE WORK

Chapter 1 introduces the concept of materials discovery and SDLs and how they can accelerate research. The work presented in this thesis demonstrates how an SDL can be made for photopolymer processing and characterization and integrated with a generative decision-making algorithm. Chapter 2 describes the physical hardware and firmware of the SDL which includes the material processing system, the curing time/kinetics system, and the automated tensile testing system which represent the processing and characterization aspects of an SDL, respectively. Chapter 3 demonstrates the integration with active learning and the acceleration of photopolymer data collection. Active learning represents the decision-making aspect of an SDL.

4.1 Contributions

The work accomplished in this thesis will help improve the SDL technology for photopolymers and for physical characterization methods. Most SDLs use non-contact methods of characterization and avoid physically manipulating the samples for testing. The results from this work can facilitate the improvement of SDL technology for photopolymers to employ more complex autonomous characterization methods.

The curing time system (CTS) is a unique high-throughput method of discovering the curing kinetics of photopolymer inks. Further, by simplifying the process, a significant portion of time can be saved when compared to other methods, such as photo-rheology.

While the data may not be as precise or provide the same amount of information, it is well suited for an active learning environment where quantity of data is important.

The automated tensile testing system (ATT) is a high-throughput method of finding photopolymer mechanical properties. To the author's best knowledge, this is the first of its kind. By using a robotic arm as the testing frame, tensile tests can be performed autonomously. Human intervention is not required for the tests and hours can be saved on data collection. Where hundreds or thousands of samples are being collected for a database, manual tensile testing is simply not feasible.

In addition, the CTS and ATT data analysis is completely automated. Every raw data output from the system is autonomously analyzed to calculate the six response properties: curing time, curing temperature, realizability, modulus of elasticity, ultimate tensile strength, and elongation at break. Also, since both systems take multiple samples, the mean and standard deviation are calculated, and outliers are removed based on 1.5 standard deviations. This value was determined experimentally to retain most of the samples and only remove true outliers. Once the outputs are discovered and cleaned, they are automatically paired with their respective input compositions of photopolymers.

Lastly, the responses collected can help support a complex active learning algorithm to predict the responses from photopolymer compositions. The SDL is integrated with an active learning algorithm to automatically receive and process data. The data can be made available to help other groups learn and predict photopolymers for DLP printing as well and help push the fore front of 3D printing materials discovery further.

4.2 Future Work

There are several future works that can be conducted to further improve the photopolymer SDL. First, and foremost, the system can be finalized to be closed loop and become a level five SDL. The breakdown in the loop is the processing of tensile bars. If that process can be automated and integrated into the existing system, then the SDL can be made much more efficient. This would require an extension to the current robot table, an additional robot that has liquid pipetting capabilities, and a machine to mold and cure the tensile bars. Figure 44 shows an example of the potential SDL improvement. Closing the loop would allow the system to run completely autonomously until the raw materials are drained. Data collection efficiency would be improved drastically as a human does not have to be present for the SDL to run, allowing it to run overnight.

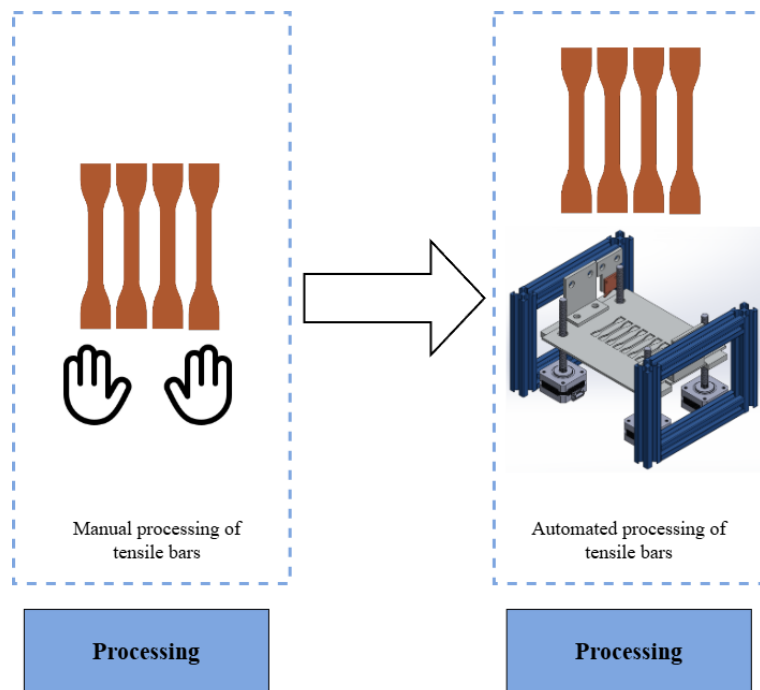


Figure 44: Closing the loop for the photopolymer SDL; automating the processing of tensile bars

Second, a real test of the reverse model can be performed to discover a photopolymer composition with targeted properties. Once enough data is collected and the active learning model becomes an accurate predictor, a case study can be performed. One example would be finding a material with skin-like properties for DLP printing.

Third, with the improvement in efficiency, the design space can be expanded to account for more potential compositions. If the crosslinker requirement was removed and each component could have four components instead of three, the total number of compositions would increase to just over 122,000. This design space would be too large to sample with the current configuration of the SDL.

Fourth, further exploration can be performed to find the extrema of the outputs in the design space. Further, it is typically bad practice to extrapolate predictions from a machine learning model, but it is unlikely that the maximum and minimum values of each response will be sampled or predictable. Therefore, optimization can be used to find the maximum and minimum possible values and create a better understanding of the design space. It would also be helpful for the model to have the extrema as the predictions will be interpolated instead of extrapolated.

Fifth and finally, additional systems can be added to the existing SDL to provide more responses. There are many other properties that are desirable to target photopolymers, including glass transition temperature (T_g) and hardness. Creating automated systems for both T_g and hardness can be helpful to better characterize the photopolymer compositions holistically.

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